



Big Data Management and Analytics

Master Thesis

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**Machine Learning and Deep Learning for Predicting
Food Security Indicators (FCS and HDDS)
From Heterogeneous Data**

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Abstract

Food security remains a critical challenge in many parts of the world, particularly in African regions where poverty, extreme weather events, socio-economic instability, and rapid population growth are major drivers of food insecurity. Despite global efforts, millions of people continue to suffer from chronic hunger and malnutrition leading to the tragic loss of lives, especially among children. As global populations grow, the demand for food increases, putting additional pressure on already strained resources. For governments and humanitarian organizations, timely and accurate identification of those suffering from food insecurity is crucial for implementing effective strategies and policies to mitigate the crisis. However, traditional methods of assessing food security, which rely on extensive surveys, are increasingly inadequate for addressing the dynamic and multifaceted nature of these challenges and sometimes not possible to be conducted in conflict-affected regions. This underscores the urgent need for more reliable, efficient, and scalable methods to monitor food security and guide interventions.

In response to these challenges, Machine Learning (ML) and Deep Learning (DL) have increasingly played a significant role in enhancing the assessment and prediction of food security. These advanced technologies offer the ability to analyze large amounts of data, providing timely and accurate insights that are critical for decision-making.

This study explores the application of ML and DL methods to predict food security indicators, specifically the Food Consumption Score (FCS) and Household Dietary Diversity Score (HDDS), using heterogeneous data sources. It focuses on the evaluation of the generalizability and robustness of the Food Security Prediction Based on Heterogeneous Data (FSPHD) framework, across three African countries Rwanda, Tanzania, and Burkina Faso.

The framework integrates diverse data types, including time-series, conjunctural and spatial, and high-resolution data, processed through models such as Long Short-Term Memory (LSTM), Random Forest (RF) and Convolutional Neural Networks (CNN) respectively. The results demonstrate the framework's effectiveness in predicting food security indicators across different geographic contexts, with the CNN model achieving superior performance in classification tasks across all studied countries when applied to high-resolution spatial data. However, challenges were observed with other models, such as RF and LSTM, which showed lower performance and variability across different countries and indicators. These findings highlight the need for further optimization of both the data and the models to enhance predictive accuracy and consistency.

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List of Abbreviations

AppEEARS	Application for Extracting and Exploring Analysis Ready Samples
CDS	Climate Data Store
CFSVA	Comprehensive Food Security and Vulnerability Analysis
CNN	Convolutional Neural Network
CSI	Coping Strategy Index
DL	Deep Learning
ERA5	ECMWF Reanalysis 5
FAO	Food and Agriculture Organization
FCS	Food Consumption Score
FDI	Foreign Direct Investment
FEWS NET	Famine Early Warning Systems Network
FIES	Food Insecurity Experience Scale
FS	Food Security
FSPHD	Food Security Prediction from Heterogeneous Data
GDP	Gross Domestic Product
GHSL	Global Human Settlement Layer
GIEWS	Global Information and Early Warning System
GMOs	Genetically Modified Organisms
HDDS	Household Dietary Diversity Score
IPC	Integrated Food Security Phase
LP DAAC	Land Processes Distributed Active Archive Center
LSMS	Living Standards Measurement Study
LSTM	Long Short-Term Memory
ML	Machine Learning
MODIS	Moderate Resolution Imaging Spectroradiometer
NASA	National Aeronautics and Space Administration
NBS	National Bureau of Statistics
NCEI	National Centers for Environmental Information
NDVI	Normalized Difference Vegetation Index
NOAA	National Oceanic and Atmospheric Administration
NISR	National Institute of Statistics of Rwanda
OSM	OpenStreetMap
RF	Random Forest
SDGs	Sustainable Development Goals
TETIS	Territoires, Environnement, Télédétection et Information Spatiale
TIFF	Tagged Image File Format
UNICEF	United Nations International Children's Emergency Fund
USAID	United States Agency for International Development
WFP	World Food Programme

Table of Contents

Abstract	i
Acknowledgments	ii
List of Abbreviations	iii
1 Introduction	1
1.1 Background	1
1.2 Literature Review	2
1.3 Objectives and Research Questions	4
1.4 Overview of the FSPHD Framework	5
1.5 Structure of the Report	6
2 Data Collection and Preprocessing	7
2.1 Study Areas	7
2.1.1 Rwanda	7
2.1.2 Tanzania	8
2.1.3 Burkina Faso	8
2.2 Data Sources	9
2.3 Ground Truth Data	10
2.3.1 Food Consumption Score (FCS)	10
2.3.2 Household Dietary Diversity Score (HDDS)	10
2.3.3 Rwanda - Groundtruth Data Collection and Preprocessing	12
2.3.4 Tanzania - Groundtruth Data Collection and Preprocessing	14
2.3.5 Burkina Faso - Groundtruth Data Collection and Preprocessing	16
2.4 Feature Data	16
2.4.1 Time-series Data	17
2.4.2 Conjunctural Data	20
2.4.3 Spatial Data	24
2.4.4 High Spatial Resolution Data	27
3 Methodology	30
3.1 General Framework	30
3.2 Feature Engineering	30
3.2.1 Normalization and Scaling	31
3.2.2 Time Window Selection	32
3.3 Model Selection, Training, and Evaluation	32
3.3.1 Model Selection	32
3.3.2 Data Splitting Techniques	33
3.3.3 Evaluation Metrics	34
4 Results and Discussion	36
4.1 Experimental Results	36
4.1.1 Regression Results	36
4.1.2 Classification Results	39
4.2 Discussion	43
5 Conclusion	45

5.1	Summary of Findings	45
5.2	Limitations	45
5.3	Future Work	46
A	Internship Progress	51
B	Illustration of Dietary Diversity Pattern	54
C	Illustration of Time series Data Distribution	55
D	Preliminary Results for Time series Data	57
E	Preliminary Results for Conjunctural and Spatial Data	60

List of Figures

1.1	The FSPHD Framework	6
2.1	Spatial Distribution of 30 Districts of Rwanda	7
2.2	Spatial Distribution 184 districts of Tanzania	8
2.3	Spatial Distribution of 351 Communes of Burkina Faso	9
2.4	Data Sources	9
2.5	FS Status of Rwanda Districts: Household Survey Data (2021)	13
2.6	Illustration of Percentile-Based Approach	13
2.7	FS Status of Tanzania Districts: Household Survey Data (2023)	15
2.8	Merging of Magharibi A and Magharib B district to Magharibi	15
2.9	Illustration of Clipping Process	17
2.10	Tanzania Rainfall Data	18
2.11	Rwanda Rainfall Data	19
2.12	Tanzania Maximum and Minimum Monthly Temperature	19
2.13	Rwanda Maximum and Minimum Monthly Temperature	20
2.14	Food Prices Trends	21
2.15	Meteorological Data Correlation	21
2.16	NDVI Data of Tanzania in 2021	25
2.17	Raster Of Land Use Land Cover Data of Rwanda	28
2.18	Raster Mean After Resampling To 100m	28
3.1	The General Framework	30
4.1	Comparison of Actual vs. Predicted FCS using Times Series Data by LSTM Model in Tanzania with different Data Splitting Techniques	37
4.2	Comparison of Actual vs. Predicted HDDS using Conjunctural and Spatial Data by RF Model for Rwanda, Tanzania, and Burkina Faso with Percentage Split	38
4.3	LSTM performance of HDDS for Rwanda with Spatio-Temporal using Time series Data	40
4.4	LSTM performance of HDDS for Tanzania with Spatio-Temporal using Time Series Data	40
4.5	LSTM performance of HDDS for Burkina Faso with Spatio-Temporal using Time series Data	40
4.6	Spatial Distribution Comparison of Actual vs. Predicted Classes of FCS by Districts for Rwanda with High Spatial Data by CNN Model using Temporal Split	41
4.7	Spatial Distribution Comparison of Actual vs. Predicted Classes of FCS by Districts for Tanzania with High Spatial Data by CNN Model using Temporal Split	42
4.8	Spatial Distribution Comparison of Actual vs. Predicted Classes of FCS for Burkina Faso by Commune with High Spatial Data by CNN Model using Temporal Split	42
B.1	Dietary Diversity Pattern of Rwanda	54

B.2	Dietary Diversity Pattern of Tanzania	54
C.1	Some of Time Series Data Distribution For Tanzania	55
C.2	Maize Data Distribution For Each Month	56
C.3	Scaled Data of Maize	56

List of Tables

2.1	Food groups and their respective weights for FCS	11
2.2	Food Groups For Household Dietary Diversity Score (HDDS)	12
2.3	Number of Household Samples Before and After Preprocessing in Rwanda	12
2.4	Number of Household Samples Before and After Preprocessing in Tanzania	14
2.5	Summary of the Feature Data with their Category	29
4.1	Comparative Regression Results of RF and LSTM Models for Timeseries Data in Rwanda, Tanzania, and Burkina Faso	36
4.2	Comparative Regression Results of CNN Model for High Spatial Resolution Data in Rwanda, Tanzania and, Burkina Faso	37
4.3	Comparative Regression Results of RF Model for Conjunctural and Spatial Data in Rwanda, Tanzania, Burkina Faso	38
4.4	Comparative Classification Results of RF and LSTM Models for Time series Data in Rwanda, Tanzania and Burkina Faso	39
4.5	Comparative Classification Results of CNN Model for High Spatial Resolution Data in Rwanda, Tanzania and Burkina Faso	41
4.6	Comparative Classification Results of RF Model for Conjunctural and Spatial Data in Burkina Faso, Rwanda, and Tanzania	42

CHAPTER 1

Introduction

1.1 Background

Food security (FS) is a multifaceted and critical issue that lies in agriculture, nutrition, health, and socio-economic development. It is defined as the state in which every individual has consistent access to an adequate amount of safe and nutritious food [2]. This notion goes beyond the availability of food to include quality, variety, and cultural experiences, ensuring the dietary needs of populations are met for an active and healthy lifestyle. The definition points to the following four (4) dimensions of food security:

1. Food Availability: Ensuring there is enough food of appropriate quality to meet dietary needs, sourced through production, imports, and aid programs.
2. Food Access: Focusing on individuals' ability to obtain food through resources and entitlements, influenced by economic, social, and legal factors.
3. Food Utilization: Emphasizing the effective use of food to meet nutritional needs and promote health, including access to clean water, sanitation, and healthcare services.
4. Food Stability: Referring to consistent access to adequate food over time, without the risk of sudden disruptions or shortages.

Major international initiatives, such as the World Food Summit in 1996 and the establishment of the Sustainable Development Goals (SDGs), particularly Goal 2: Zero Hunger, have emphasized the global commitment to eradicating hunger and ensuring food security for all [3, 4, 5]. These initiatives have led to numerous advancements, including improved agricultural practices, enhanced food distribution systems, and better monitoring of food security worldwide.

Despite these initiatives, food insecurity remains a significant challenge globally, affecting millions of people and resulting in severe consequences. For example, malnutrition, as a direct consequence of food insecurity, has become a leading cause of child mortality worldwide. Approximately 5 million children under the age of five die annually due to malnutrition and related health issues [6]. In 2023, the United Nations International Children's Emergency Fund (UNICEF) reported that around 181 million children under five suffer from severe food poverty, making them up to 50% more likely to experience wasting, a life-threatening form of malnutrition[7]. Recent statistics further highlight the extent of the situation, with around 258 million people across 58 countries facing acute food insecurity in 2022, a substantial rise from 193 million in 2021.[8]

Early warning systems like the Global Information and Early Warning System (GIEWS) managed by FAO, and the Famine Early Warning Systems Network (FEWSNET) established by the United States Agency for International Development (USAID) play a crucial role in monitoring global food security. These systems utilize advanced data analytics to enhance their

monitoring capabilities. For instance, FEWSNET integrates various data sources, including livelihoods, agroclimatology, markets, trade, and nutrition data, to forecast food security conditions [9]. Similarly, GIEWS employs satellite imagery and economic and agricultural data to continuously assess food security worldwide, providing early warnings of potential food shortages[10]. Although these systems provide valuable insights, they have limitations. They rely on a limited set of data, often excluding direct measures of food consumption at the household level. Additionally, while they capture key environmental and economic indicators, they may not fully account for factors such as local market dynamics, household-level dietary diversity, and coping mechanisms, which are crucial for a comprehensive understanding of food security on the ground.

To address these gaps and provide a more detailed understanding of food security at the household level, organizations such as the World Food Programme (WFP) and the Food and Agriculture Organization (FAO) play a crucial role in monitoring food insecurity worldwide. They establish different food security indicators such as the Food Consumption Score (FCS), Household Dietary Diversity Score (HDDS), Food Price Index, Food Insecurity Experience Scale (FIES), reduced Coping Strategy Index (CSI), and others to monitor and assess food security, identifying regions in need and implementing targeted interventions. However, these indicators are traditionally calculated and identified through comprehensive household surveys which can be resource-intensive, time-consuming, and prone to reporting biases. Furthermore, in certain areas, particularly those affected by conflict, disease, or geographic remoteness, conducting these surveys may be impossible.

As the global population continues to grow, projected to reach nearly 10 billion by 2050 [11], ensuring food security for all becomes increasingly critical. Moreover, the challenges of climate change, economic instability, and political conflicts highlight the urgent need for more reliable and faster methods to monitor food insecurity. These traditional survey-based methods may not be able to keep up with the dynamic nature of food insecurity, where rapid changes in economic conditions or environmental factors can quickly alter the food security landscape. Delays in data collection and analysis can lead to delayed responses, worsening the suffering of those affected.

To overcome the limitations of traditionally based survey methods, there is a growing interest in utilizing advanced machine learning (ML) and deep learning (DL) techniques to develop systems that provide timely and accurate information, thereby enhancing the efficiency and accuracy of food security assessments. Recent research highlights that ML and DL have been increasingly applied in predicting food security indicators directly from the drivers associated with food security. This growth reflects a significant shift towards data-driven approaches in food security assessment. However, there remain gaps in understanding the full potential and limitations of these methods, particularly across different geographic regions and socio-economic contexts [12],[1]. In the next section of the literature review, we will examine these studies in detail, focusing on the methodologies employed, the types of data utilized, and the effectiveness of ML and DL in predicting various food security indicators and identify areas where further research is needed to enhance the predictive accuracy and applicability of these models in diverse settings.

1.2 Literature Review

In recent years, research has significantly increased utilizing machine and deep learning methodologies for monitoring food security. Numerous studies have demonstrated the potential of these methods to analyze and integrate various heterogeneous data drivers to estimate traditional indicators such as the Integrated Food Security Phase (IPC), FCS, HDDS,rCSI)

These data cover environmental factors such as temperature, evaporation, rainfall, and vegetation indices such as the (Normalized Difference Vegetation Index (NDVI) for instance) derived from remote sensing data. Structural factors, which remain constant over time, include population count, land use cover, and soil type. Additionally, some research incorporates economic data, commodity prices, conflict data [12], [1], [13], [14].

These data serve as critical drivers of FS, influencing factors such as agricultural productivity, market stability, and the socio-economic resilience of population classes. To propose effective FS strategies, it is crucial to consider the role of heterogeneous data across these interconnected dimensions. For instance, [15] demonstrated the use of open data such as the NDVI, food prices, climate data, market, infrastructure, demography variables, and livelihood zone characteristics with a Random Forest (RF) and Gradient Boosting (GB) algorithms for the prediction of 3 IPC classes and IPC changes in Ethiopia. Their study shows that IPC scores can be estimated for the classification with an accuracy of 90%. Interestingly, the IPC future changes can be better forecasted when forecasting long-term (7 months; $F1 = 0.61$) compared to short-term (3 months; $F1 = 0.51$). The most relevant predictors were reported to be FS history and surface soil moisture. Even though IPC incorporates consensus-based expert opinion and global scientific standards in its food insecurity evaluations, it is not available at the sub-national district level.

Another study [13] proposed the prediction of three FS indicators (rCSI, HDDS, FCS) through linear and log-linear models. Their model incorporated spatially and temporally granular market data, remotely sensed rainfall, geographic data, and demographic characteristics. The model was trained on 2010 data in Malawi to forecast FS outcomes in 2013. They are successfully identifying the FS status of 83 to 99% of the most food-insecure village clusters. However, the model faced limitations due to the varying availability and quality of data, as well as challenges in predicting FS in regions with different agroecological and market conditions.

Building on these advancements, more sophisticated approaches have been developed to integrate multiple data sources and improve predictive accuracy at a larger scale. The research by [14] presents a method to predict food insecurity at the first-order administrative level by integrating secondary data on key drivers such as conflict, economic shocks, and extreme weather events. The data drivers include prices of staples, headline and food inflation, currency exchange rates, and gross domestic product (GDP) per capita) and data on rainfall and vegetation. The approach employs GB regression trees (XGBoost) to predict FCS and the rCSI. The models explain up to 81% of the variation in insufficient food consumption and 73% of the variation in crisis or above food-based coping levels. Data from 78 countries for food consumption and 41 countries for food-based coping strategies were used in the models. Despite these strengths, the models face limitations due to the combination of sub-national data from various countries, which may overlook local-specific patterns.

In this context, the research by [1] which forms as a basis for our study introduces the Food Security Prediction based on Heterogeneous Data (FSPHD) framework, which builds on these concepts by combining ML such as Random Forest (RF) and DL (such Convolutional Neural Networks (CNN) and Long short-term memory (LSTM)) methods to predict FCS and HDDS. This study integrates a wide range of publicly available heterogeneous data, including rasters, GPS points, line vectors, quantitative variables, and time series using different data fusion techniques. Its innovative use of data fusion to integrate these heterogeneous sources for predictions sets a new standard for future research, crucial for addressing the multifaceted challenges of food insecurity. The experimental results based on data from Burkina Faso between 2009 and 2018, demonstrate the framework’s promising performance, achieving an R^2 of 0.469 for FCS and 0.434 for HDDS. The limitations in this study include the performance of the models being

highly dependent on the specific data used, and may not generalize well to other contexts, the lack of interpretability of the neural network models, and the lower performance of the models on the time series data.

All of these studies highlight that ML and DL have the potential to estimate food insecurity and can be useful for governments and humanitarian organizations to make informed and timely decisions, offering innovative solutions. However, despite the contributions of ML in this field, several challenges need to be addressed in future research. [12] provides a systematic review of various ML and DL models used for food security tasks and identifies significant challenges. One major challenge is the heterogeneity and complexity of food security data from diverse sources, which require extensive preprocessing to ensure compatibility and accuracy. Additionally, variability in data availability and quality can complicate integration efforts and risk introducing biases into model outputs, emphasizing the need for high-quality data, as noted by [13] and [14]. Another challenge highlighted by [12] is the high computational demands of processing large and complex datasets, particularly in DL models, which are resource-intensive. Furthermore, the "black box" nature of DL models can limit interpretability, hindering the understanding of the decision-making process, as discussed by [1]. Moreover, a critical challenge in developing predictive models for food security is ensuring their generalizability across different contexts [1][12].

Food security data can vary significantly depending on the region's specific socio-economic, climatic, and agricultural conditions. This variability makes it difficult for models trained on data from one region to accurately predict outcomes in another, as the underlying factors influencing food security can differ widely. Addressing this challenge requires developing adaptable models that can incorporate and adjust to diverse datasets, ensuring they remain relevant and accurate across various contexts.

1.3 Objectives and Research Questions

The primary objective of our work is to evaluate and extend the FSPHD Framework to enhance its generalizability and robustness for predicting FCS and HDDS in other African countries. This objective directly addresses the challenge identified in the literature review concerning the need for models to be adaptable to diverse geographic and socio-economic conditions. Specifically, we aim to:

- **Adapt the FSPHD framework to different contexts:** We seek to determine how the framework can be modified to be applicable in diverse geographic and socio-economic settings.
- **Improve data adaptation processes:** This involves, data collection and preprocessing, for example computing FCS and HDDS from household survey data, developing methods to handle varying data types and qualities across different regions.
- **Utilize Advanced ML and DL Techniques:** We aim to incorporate advancements ML and DL techniques to improve the accuracy and reliability of the FSPHD Framework.
- **Extend the framework For regression and classification:** We aim to enable the FSPHD to predict various aspects of food security.

To achieve these objectives we propose the following key research questions that will guide our work by focusing on key aspects necessary for the successful adaptation and optimization of the FSPHD Framework

- **RQ1: Can the FSPHD framework be generalized for other African countries (e.g., Rwanda and Tanzania) beyond Burkina Faso?**

This question focuses on exploring the scalability and applicability of the FSPHD framework in different countries. It would lead to an investigation of the framework’s flexibility, robustness, and overall design to handle diverse environments.

- **RQ2: Which specific data adaptations are needed to apply within the FSPHD framework for other African countries with different climatic and socio-economic conditions?**

This question focuses on identifying specific data modifications, and adjustments in data processing such as different normalization techniques, and integration methods to ensure the framework’s effectiveness in new contexts.

- **RQ3: Which aspects of ML and DL techniques can be optimized within the FSPHD framework to improve performance in predicting food security indicators?**

This question aims to explore the optimization of ML and DL techniques within the FSPHD framework to identify specific techniques such as hyperparameter tuning to enhance the framework’s predictive accuracy and reliability when applied to diverse datasets from different regions.

- **RQ4: How can the FSPHD framework be adapted to incorporate both regression and classification tasks effectively?**

This question examines the framework’s capability to handle both regression and classification models, assessing its effectiveness in predicting different aspects of food security. This dual capability of predicting both regression and classification provides flexibility to handle either or both tasks as needed in future applications

1.4 Overview of the FSPHD Framework

The FSPHD framework integrates various heterogeneous data sources and ML techniques to predict food security indicators. The framework takes different types of input data with different granularities. These input data include time series data, conjunctural and spatial data, and high spatial resolution (HSR) data.

Different data fusion techniques were tested for integrating these heterogeneous data sources. These techniques included feature fusion, where features extracted by the deep learning models are combined, and decision-level fusion, where predictions from different models are aggregated. In the feature fusion approach, data such as time series (processed using Long Short-Term Memory (LSTM) networks) and high spatial resolution data (handled by Convolutional Neural Networks (CNNs)) are integrated alongside conjunctural data and spatial data. These integrated features are then fed into a Random Forest (RF) model to generate predictions for food security indicators such as the Food Consumption Score (FCS) and Household Dietary Diversity Score (HDDS). In contrast, the decision-level fusion branch aggregates the predictions from the LSTM, CNN, and RF models using a linear model with ridge regularization. While both approaches were explored, the feature fusion branch demonstrated superior performance. The diagram presented in Figure 1.1 is based on this branch that yielded the best results, utilizing feature fusion to create new data representations that better correlate with the response variables. In this study, we will focus on evaluating this feature fusion branch, given its superior performance in preliminary assessments.

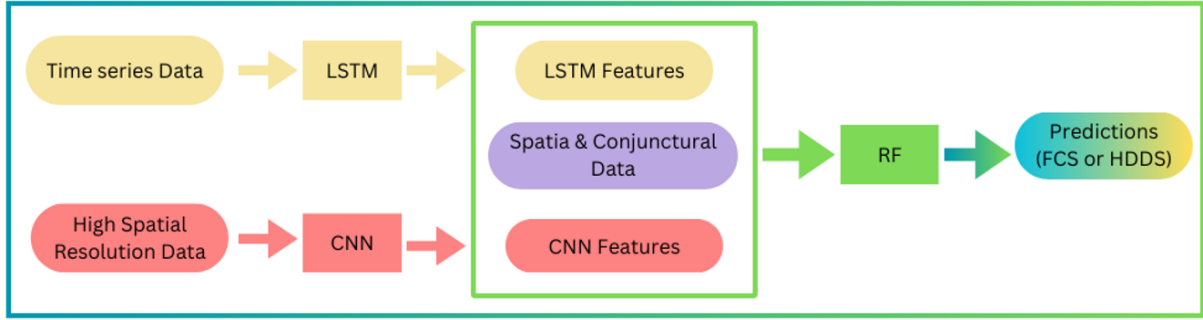


Figure 1.1: The FSPHD Framework

1.5 Structure of the Report

This report is organized into **five (5)** main chapters, each addressing different aspects of the research. **Chapter 1** provides an introduction, including background, an overview of the ES-PHD Framework, related work, and research objectives. **Chapter 2** discusses data collection and preprocessing, detailing the study areas, data sources including ground truth data, and feature data. **Chapter 3** covers methodology, including the general overview of the framework, feature engineering, model selection, training, and evaluation. **Chapter 4** presents the results and discussion, including experimental results and discussions. Finally, **Chapter 5** concludes the report with a summary of findings, limitations, and future work.

CHAPTER 2

Data Collection and Preprocessing

2.1 Study Areas

This study will focus on two East African countries; Rwanda, and Tanzania, and one West African country; Burkina Faso. These countries offer diverse climatic conditions, agricultural practices, and socio-economic landscapes, providing a comprehensive testing ground for the FSPHD framework. By examining the framework's performance in these varied settings, we aim to identify and address specific regional challenges, ensuring that the framework is adaptable and effective across different countries, particularly in Africa.

2.1.1 Rwanda

Rwanda is a small landlocked country located in the heart of East Africa. Agriculture forms the backbone of Rwanda's economy, employing a large portion of the population and contributing significantly to GDP [16, 17]. The country is divided into four main administrative levels: Provinces, Districts, Sectors, and Cells. There are five (5) Provinces which are further divided into 30 Districts as shown in Figure 2.1, Districts are subdivided into Sectors, and the Sectors into Cells, which are the smallest administrative units. The country is vulnerable to climate change, experiencing erratic rainfall patterns, prolonged droughts, and extreme weather events. These environmental factors disrupt agricultural productivity, leading to crop failures and food shortages [18].

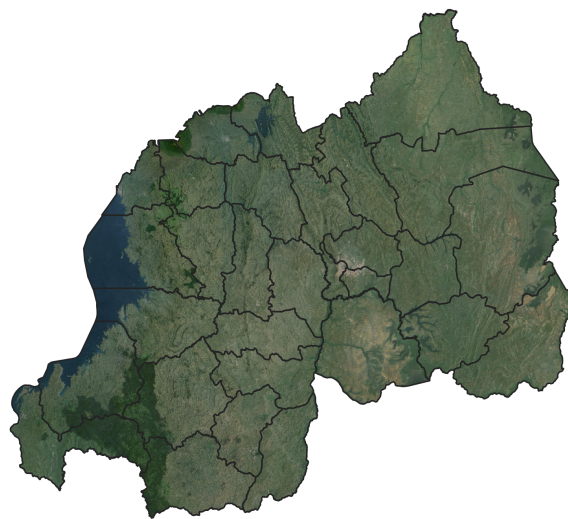


Figure 2.1: Spatial Distribution of 30 Districts of Rwanda

2.1.2 Tanzania

Tanzania is a country located in East Africa, and its economy is primarily driven by agriculture. The country is administratively divided into 31 regions, further subdivided into 184 districts as shown in Figure 2.2. Each district is further divided into wards which is the lowest administrative level. Despite being an agricultural country, Tanzania faces challenges with malnutrition and lack of dietary diversity, particularly in rural areas[19]. Many households rely heavily on staple crops such as maize, rice, and cassava, leading to a limited variety of diets and micronutrient deficiencies. Also, poverty remains a significant barrier that makes many households struggle to afford an adequate and nutritious diet[20]. Apart from that, limited access to markets, infrastructure, and transportation influences the issues of food insecurity, particularly in remote and rural areas.

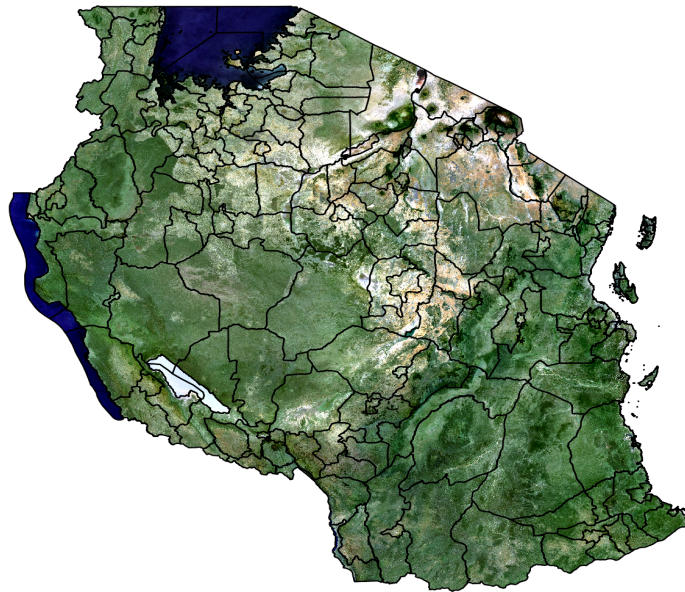


Figure 2.2: Spatial Distribution 184 districts of Tanzania

2.1.3 Burkina Faso

Burkina Faso is a landlocked country in West Africa with an economy heavily dependent on agriculture and mining. Administratively, Burkina Faso is divided into 13 regions, further subdivided into 45 provinces. Each province is further divided into 351 communes as shown in Figure 2.3, serving as the smallest administrative units. It is a low-income Sahelian country with limited natural resources, and more than 40% of the population lives below the national poverty line, making it vulnerable to food insecurity [21]. In 2024, 2.7 million people are reported to experience acute food insecurity a substantial rise from approximately 700,000 in 2019. The climate changes, political instability, ongoing armed conflicts, climate changes, dependent on rain-fed agriculture and lack of infrastructure are the main contributors to food insecurity in Burkina Faso [22]



Figure 2.3: Spatial Distribution of 351 Communes of Burkina Faso

2.2 Data Sources

The primary objective of this study is to predict food security indicators at the district for Rwanda and Tanzania and commune level for Burkina Faso. The dataset used is divided into ground truth data and feature data. Ground truth data includes FCS and HDDS which serve as a responsive variable for our predictions. Feature data consists of various datasets utilized for predicting food security indicators. This includes land use land cover data, meteorological data, food prices data, economic data, elevation data, NDVI data, population data, waterways data, and others as illustrated in Figure 2.4. These features data are then integrated into four groups: Time-series Data, Conjunctural Data, Spatial and High Spatial Data.

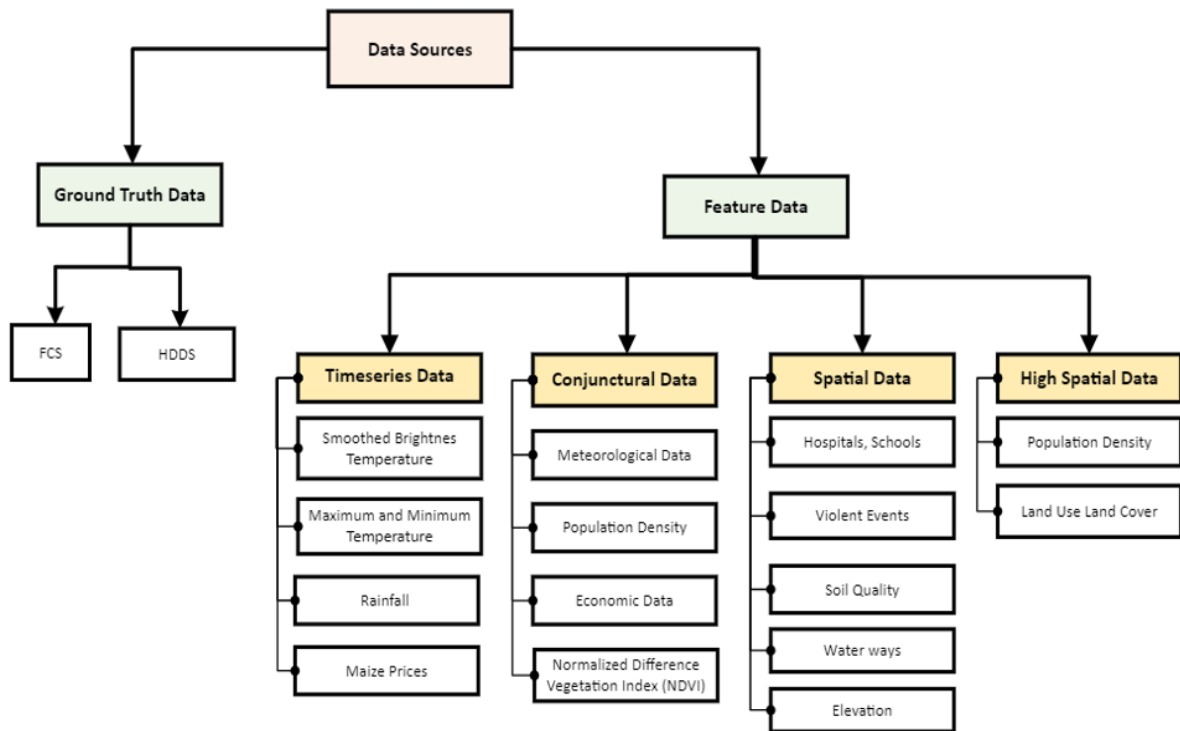


Figure 2.4: Data Sources

2.3 Ground Truth Data

FCS and the HDDS are food security indicators derived from Household Survey Data. These food security indicators provide valuable insights into the dietary patterns and nutritional status of households used for assessing food security. According to the dimensions of food security - food availability, food access, food utilization, and food stability, as outlined by the FAO [2] the FCS addresses both food access and food utilization, while the HDDS focuses primarily on food access. In the following subsections, we will explore the calculation of these indicators and their alignment with those dimensions of food security.

2.3.1 Food Consumption Score (FCS)

FCS is a food security indicator developed by the World Food Programme(WFP) [23] that assesses dietary diversity, food frequency, and the relative nutritional importance of different food groups. It is calculated based on the frequency of consumption of different food groups consumed by a household during the past 7 days. By reflecting nutritional quality and dietary adequacy through the diversity and frequency of food groups consumed it can assess food utilization which also indicates food access as a higher FCS suggests the better economic and physical ability of households to be able to access diverse and rich nutritious foods.

The calculation of the FCS involves a systematic approach to evaluate dietary diversity and frequency. First, food items are categorized into predefined food groups. For each food group, the consumption frequencies of all items within that group are summed, with a maximum cap of 7 days to prevent overestimation. Each food group's total frequency is then multiplied by a specific weight that reflects the relative nutritional value of that group, these weights are defined by WFP as outlined in Table 2.1. These weighted scores are then summed to derive the overall FCS. Upon obtaining the FCS, it is crucial to establish thresholds for Food Consumption Groups (FCGs) to interpret the score and categorize households into different levels of food security. The WFP [23] suggests the three thresholds for FCGs which are [0 - 21], [21.5 - 35], and [≥ 35] representing poor, borderline, and acceptance respectively. However, it is suggested to assess these thresholds and potentially adjust them according to the context and dietary habits of the population under consideration so that it will precisely reflect the chosen context. The WFP [23] provides a guideline on how the new thresholds can be set by considering different scenarios such as the regions with high consumption of sugar and oil. Mathematically, the FCS is calculated as:

$$FCS = \sum_{j=1}^9 f_j \times x_j \quad (2.1)$$

where f_j represents the frequency of consumption of a food group j , x_j represents the nutritional value (weight) of a food group j , and j ranges from 1 to 9, representing the nine food groups considered in the calculation.

2.3.2 Household Dietary Diversity Score (HDDS)

The HDDS is a qualitative measure of food consumption developed by the Food and Nutrition Technical Assistance (FANTA) project that focuses on the variety of foods consumed in the diet for the past 24 hours. Since it does not capture the frequency or nutritional weight it

SN	Food Items	Food Groups	Weights
1	Maize, rice, sorghum, bread, other cereals, cassava, potatoes, sweet potatoes and other tubers	Cereals and Tubers	2
2	Beans, peas, groundnuts, and cashew nuts	Pulses and Nuts	3
3	Vegetables such as carrots, cucumbers, tomatoes, leaves such as cassava leaves, cabbage	Vegetables	1
4	Fruits such as oranges, limes, apples, grapes, strawberries, and similar	Fruits	1
5	Beef, goat, poultry, pork, eggs, and fish	Animal Protein	4
6	Milk, yogurt, and other dairy	Product	4
7	Sugar, sugar products, honey	Sugar	0.5
8	Oils, fats, and butter	Oil	0.5
9	Spices, tea, coffee, salt	Condiments	0

Table 2.1: Food groups and their respective weights for FCS

can assess only the dimensions of food access. The calculation of HDDS consists of a simple count of food groups that a household has consumed over the past 24 hours. Each food group is then assigned a score of 1 if consumed over the past 24 hours otherwise 0. The household score will range between 0 to 12 and is equal to the total number of food groups consumed by the household. These 12 food groups used for HDDS are represented in Table 2.2. Unlike the FCS, there are no universally accepted thresholds for HDDS that directly categorize households into food security groups. Therefore, when using HDDS in food security analysis, it is essential to consider the context and local dietary practices and it should also be complemented with other indicators such as FCS to provide a clear picture of food security conditions. However, A higher HDDS indicates a more varied and potentially more nutritious diet, while a lower HDDS suggests limited dietary diversity and potential nutritional inadequacies. Mathematically, the HDDS is calculated as:

$$HDDS = \sum_{j=1}^{12} x_j \quad (2.2)$$

where x_j equals 1 if the household consumed food from group j in the past 24 hours, and 0 otherwise.

It is important to note that, in our case, the HDDS was computed using only nine (9) food groups, the same as those used for FCS for both countries; Rwanda and Tanzania. This decision was made because most of the survey data did not include the detailed information necessary for computing HDDS based on the consumption of food over the past 24 hours. Therefore, we decided to use the same data used for computing the FCS to compute the HDDS. While this approach ensures consistency across our datasets, it may also represent a limitation of our study, as we could not obtain the specific data required for a traditional HDDS calculation. However, given the constraints, this was the most feasible solution.

SN	FOOD GROUP
1	Cereals
2	Roots and tubers
3	Vegetables
4	Fruits
5	Meat, poultry, offal
6	Eggs
7	Fish and seafood
8	Pulses/legumes/nuts
9	Milk and milk products
10	Oils/fats
11	Sugar/honey
12	Miscellaneous

Table 2.2: Food Groups For Household Dietary Diversity Score (HDDS)

2.3.3 Rwanda - Groundtruth Data Collection and Preprocessing

Household survey data from Rwanda was provided by the National Institute of Statistics of Rwanda (NISR) every three years from 2006 to 2021. These publicly available datasets include varying numbers of household samples each year as represented in Table 2.3, reflecting changes in survey scope and coverage. The data were collected by directly downloaded from the NISR website however, data from the year 2009 was not used in this study due to missing important information for computing the FCS and HDDS.

Year	Household Samples	Samples After Preprocessing
2006 ^[24]	2,783	2,769
2012 ^[25]	7,498	7,496
2015 ^[26]	7,500	7,498
2018 ^[27]	9,709	9,708
2021 ^[28]	9,000	9,000

Table 2.3: Number of Household Samples Before and After Preprocessing in Rwanda

After data collection, the data was preprocessed to prepare for computing the FCS and HDDS. Samples with incomplete information were excluded during preprocessing. The scores were computed on an individual household basis and then aggregated by mean to the district level, as the focus of this study is to evaluate food security at the district level rather than for individual households. To ensure the accuracy of spatial analysis, it was necessary to reconcile the administrative names in the household data with the corresponding administrative boundaries in the shapefiles. This step involved matching the district names in the survey data to those in the geographic data, ensuring consistency and alignment between the datasets.

Next, the FCS groups were initially assigned using the default thresholds recommended by WFP. However, the resulting classes were not normally distributed, with some classes having significantly more data than others as shown in Figure 2.5(a) where the number of acceptance classes (in green color) dominated in many districts. To address this issue, we use WFP guidelines as illustrated in the appendix B to find the appropriate thresholds for each country, however, the dietary pattern shows that the default thresholds aligned with the context. Therefore to solve this problem, a percentile-based approach was employed to set new FCS thresholds which are

[< 42], [42 - 47], and [$\geq$ 47] representing poor, borderline and acceptance respectively. This method utilized data percentiles to create more balanced class distributions as shown in Figure 2.5(b) where the classes appear more evenly distributed. While this approach effectively addressed the class imbalance, it introduced a limitation, as the new thresholds did not perfectly align with the standard WFP thresholds, potentially affecting the accuracy and comparability of food security predictions.

Since HDDS does not have any default threshold for food security level we use the same method of a percentile as we have applied in FCS to set the classes and the threshold values of [< 4.8], [4.8 - 5.5), and [≥ 5.5] was obtained representing poor, borderline and acceptance respectively.

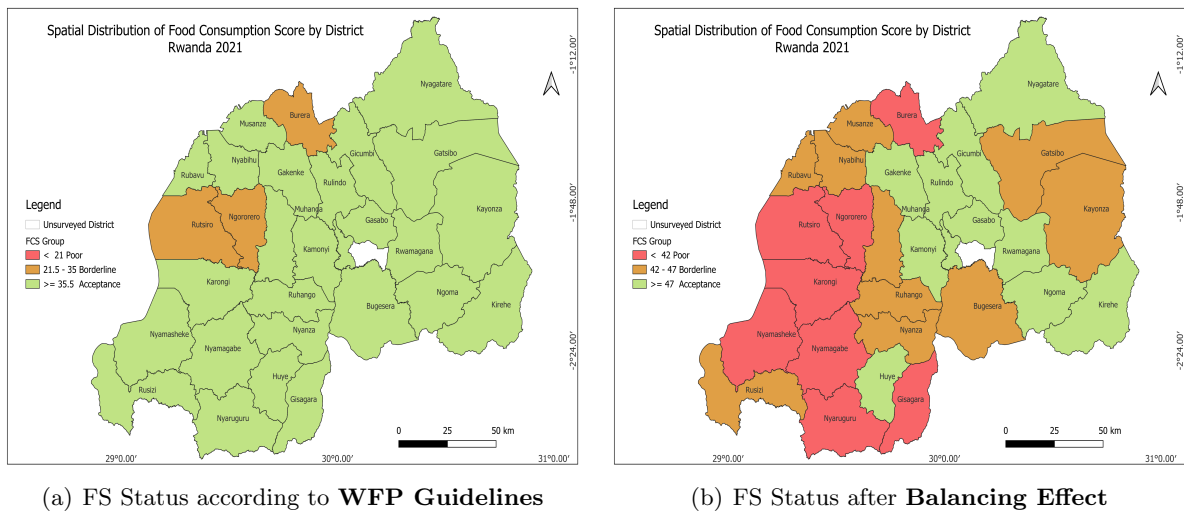


Figure 2.5: FS Status of Rwanda Districts: Household Survey Data (2021)

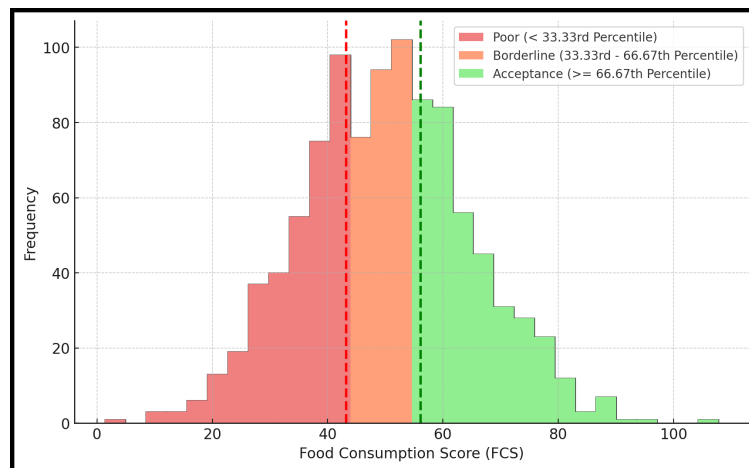


Figure 2.6: Illustration of Percentile-Based Approach

Figure 2.6 illustrates the distribution of FCS values and the classification into three categories: poor, borderline, and acceptance. The poor class includes FCS values below the 33.33rd percentile, the borderline class spans from the 33.33rd to the 66.67th percentile, and the acceptance class includes values at or above the 66.67th percentile. The red and green dashed lines mark these percentile thresholds, clearly describing the boundaries between classes. The main reason

for using this method is to achieve more balanced classes, which is crucial for classification tasks, ensuring that each class contains a similar proportion of data.

2.3.4 Tanzania - Groundtruth Data Collection and Preprocessing

Household survey data for Tanzania was collected by the National Bureau of Statistics (NBS) and is available through the World Bank’s Living Standards Measurement Study (LSMS). These surveys, conducted over multiple years, provide comprehensive information on household characteristics, including food consumption, dietary habits, demographic details, and socio-economic factors. This data consists of 6 datasets that have different numbers of household samples as represented in Table 2.4. However, the survey data presents some challenges, particularly due to how data collection was conducted across two different years. For example, from August 2010 to August 2011, data collection within the same districts was done across two years. This overlap created challenges in data aggregation, as some samples from the same district were collected in 2010 and others in 2011. Treating each year separately would lead to an imbalance in sample sizes, with one year potentially having significantly more data than the other, which could negatively impact the model’s training and lead to biased results. But also if we use the way it is it would create a challenge in handling time series data. To address this issue we opted to use the last year $y + 1$ as the reference year and included data from year y within a time window framework. This approach ensures consistency in our temporal data and allows us to utilize the complete dataset, thereby providing a balanced and comprehensive dataset for more accurate and robust model training.

Year	Household Samples	Samples After Preprocessing
2010/2011 ^[29]	3,924	3,841
2012/2013 ^[30]	5,010	4,921
2014/2015 ^[31]	3,352	3,330
2019/2020 ^[32]	1,184	1,136
2020/2022 ^[33]	4,709	4,572
2021/2024 ^[34]	1,049	1,048

Table 2.4: Number of Household Samples Before and After Preprocessing in Tanzania

These data were processed using the same methodology to calculate the FCS and HDDS as the data from Rwanda. However, similar issues with class imbalance were observed, with many districts falling into the acceptance food security group. To address this, a percentile-based approach was applied, resulting in new FCS thresholds of [<50.5], [$50.5 - 59.5$), and [≥ 59.5] which represent poor, borderline, and acceptance respectively. Figure 2.7(a) shows the FCS distribution using the default thresholds recommended by WFP and the Figure 2.7(b) displays the FCS distribution with more balanced classes. Similarly, the percentile-based approach was applied to get HDDS thresholds [< 6.8], [$6.8 - 7.5$), and [≥ 7.5] representing poor, borderline, and acceptance respectively.

Apart from the class imbalance issues in Tanzania, another challenge was encountered during the reconciliation process of district names. Specifically, inconsistencies were identified between the districts listed in the survey data and those in the administrative boundary shapefiles for Tanzania. These inconsistencies likely arose from ongoing changes in administrative boundaries within Tanzania, which were not reflected in the outdated shapefiles. As a result, some districts listed in the survey data were missing from the shapefiles. To address this, the existing districts

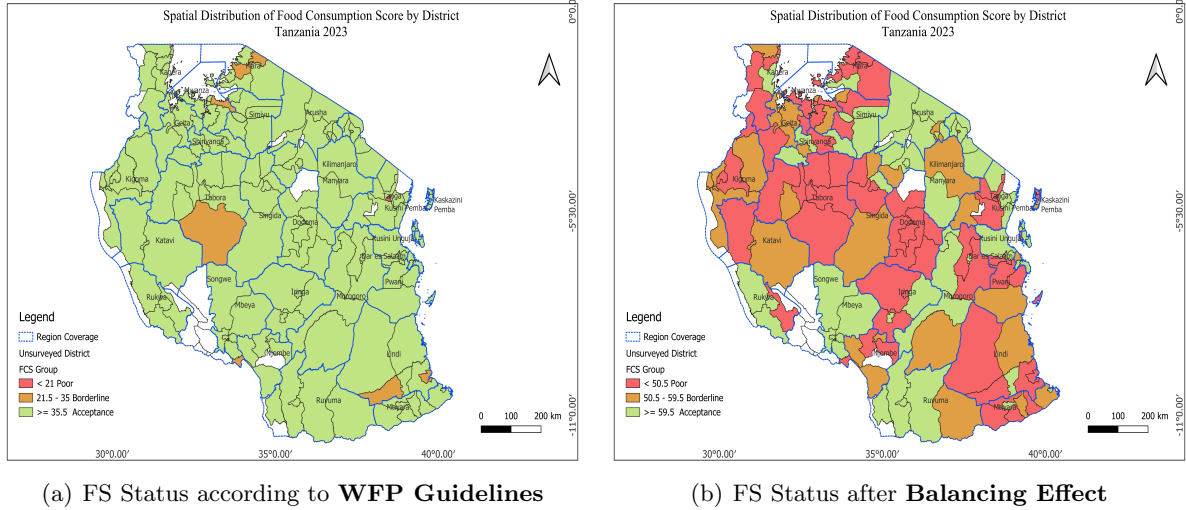


Figure 2.7: FS Status of Tanzania Districts: Household Survey Data (2023)

in the shapefiles were cross-referenced with both the old and new districts identified in the survey data. Districts that had been split or combined were merged to reflect the same boundaries as in the survey data. This approach ensured that the survey data could be properly used without losing valuable information due to administrative changes.

One specific example is the district of Magharibi, which is split into Magharibi A and Magharibi B in the shapefiles but listed as a single district in all survey datasets. Since there was no additional information to distinguish between Magharibi A and Magharibi B in the survey data, these two districts were merged into a single feature named Magharibi in the shapefiles as shown in Figure 2.8(a) and Figure 2.8(b) representing before and after the merge respectively. By modifying the shapefiles to reflect these reconciliations, consistency across the data was ensured. This updated shapefile will be used for all subsequent analyses in the study.

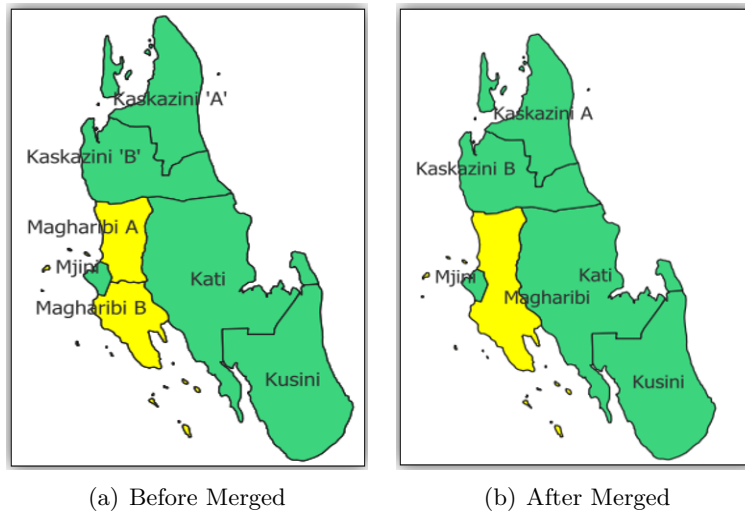


Figure 2.8: Merging of Magharibi A and Magharib B district to Magharibi

2.3.5 Burkina Faso - Groundtruth Data Collection and Preprocessing

The ground truth data for Burkina Faso were privately provided by the Burkinabe Ministry of Agriculture to TETIS for the development of the FSPHD framework. This dataset includes agricultural survey data from 2009 to 2018, covering 46,400 farm household samples across 344 of the 351 communes per year [1]. In our study this data was provided by TETIS since we aim to generalize the FSPHD, it was crucial to have this data so that we can test our new models across all selected countries i.e (Burkina Faso, Rwanda, and Tanzania). The FCS and HDDS were already calculated, with the data aggregated at the commune level. Further details on the data processing can be found in the research by [1].

We further preprocessed the dataset by defining the FCS and HDDS classes. We encountered class imbalance issues similar to those in Tanzania and Rwanda when using the default WFP thresholds. To address this, we applied the percentile method to set new thresholds. The new thresholds for FCG become $[<48.8]$, $[48.8 - 58.9)$, and $[>=58.9]$ which represent poor, borderline, and acceptance respectively. For HDDS are $[<4.9]$, $[4.9 - 5.9)$, and $[>=5.9]$ representing poor, borderline, and acceptance. Additionally, we reconciled the names of administrative boundaries, particularly commune and province names, in the shapefiles with those in the ground truth data file.

2.4 Feature Data

The feature data include diverse datasets essential for predicting food security indicators, such as meteorological data, land use and land cover data, population density, elevation, economic, and more as shown in Figure 2.4. These data are categorized into four groups: time series data, spatial data, conjunctural data, and high spatial resolution data. The collection and categorization of these datasets is grounded in the methodology established by [1] during the development of the FSPHD framework. Since this study aims to evaluate the generalizability of the FSPHD framework to new study areas, it was essential to collect the same categories of data to allow for a consistent and comprehensive evaluation of the model's performance in these new study areas. This approach will help to address Research Question 1 (*Can the FSPHD framework be generalized for other African countries (e.g., Rwanda and Tanzania) beyond Burkina Faso?*).

Since these data come from different sources and have different structures, each dataset initially undergoes individual preprocessing to ensure consistency and compatibility before being integrated into the specified categories. This involves data curation and transforming the data into a common format suitable for each category. Given that the study focuses on predictions of the FCS and HDDS at the district level, all feature data with finer granularity, such as temperature and rainfall, were aggregated to the district level. Economic data were aggregated at the country level using different aggregation methods depending on the nature of the data, such as summing for rainfall and averaging for temperature. Detailed descriptions of the resolutions, temporal resolution, sources, and scaling-up methods used for each dataset will be provided in the category section of each data source, with a summary provided in Table 2.5.

Most of the feature data used in this study are derived from satellite sources and are available in formats such as NetCDF or TIFF. These data are initially extracted on a country-wide basis. To ensure the data accurately represent each district, a clipping method was employed. Clipping as illustrated in Figure 2.9 is a geoprocessing operation that extracts data within specified geographic boundaries, effectively isolating the data for the area of interest. This method

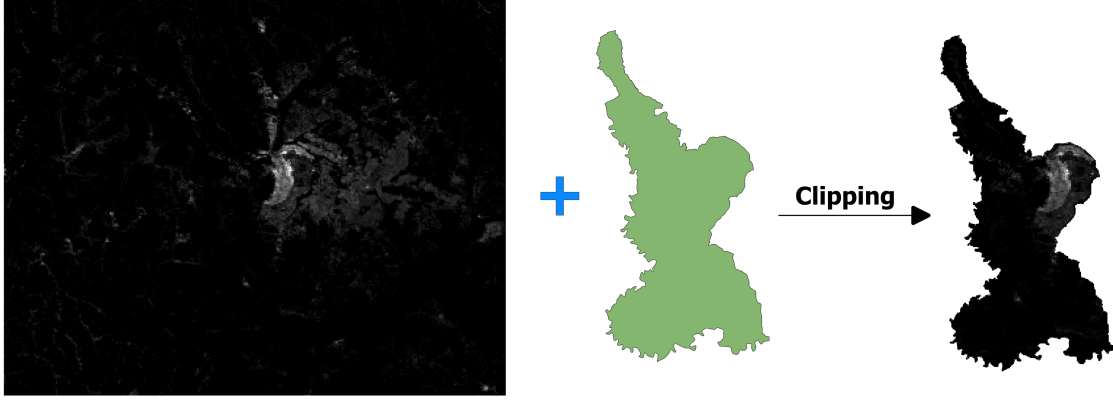


Figure 2.9: Illustration of Clipping Process

involves using the boundary of each district to ensure that the data values are specific to these defined areas, maintaining high-quality and contextually relevant information for predicting food security indicators at the district level.

It is important to note that this section only covers the collection and preprocessing of feature data for Rwanda and Tanzania. The feature data for Burkina Faso were provided by TETIS, and they had already been preprocessed and prepared for model training.

2.4.1 Time-series Data

This category includes data with multiple values per year, recorded monthly at the district level, for both Rwanda and Tanzania. It comprises datasets such as rainfall, temperature, and maize prices, among others. Each of these data types provides valuable insights into different factors that affect food production, availability, and market dynamics, ultimately impacting food security. For example, rainfall patterns influence crop yields, temperature variations affect growing conditions, and maize prices reflect market stability and access to food.

Smooth Brightness Temperature: The Smooth Brightness Temperature data, provided by the National Oceanic and Atmospheric Administration (NOAA) [35], measures the radiance emitted by the Earth’s surface as detected by satellites. This brightness temperature data represents the thermal characteristics of the land surface, which helps in understanding factors such as vegetation cover, soil moisture, and land use, and their impact on agricultural conditions. High brightness temperatures can indicate heat stress or drought conditions, which negatively affect crop yields, while lower brightness temperatures might suggest healthy vegetation with sufficient moisture.

This data is collected and extracted via Google Earth Engine and is originally in TIFF file format, with a spatial resolution of 0.07 degrees. The data is then aggregated for each district by taking the maximum monthly temperature to capture the highest temperature recorded each month. Initially measured in Kelvin, the temperature values are converted to Celsius for standardization and easier interpretation.

Rainfall Data: Rainfall data is collected from the ERA5 dataset [36] using the API provided by the Copernicus Climate Data Store (CDS) [37]. The ERA5 dataset offers high-resolution

climate data at a 0.25° spatial resolution. The data is initially collected in NetCDF format, covering the entire country, and then the clipping method is applied to extract data of each district. In preprocessing, the monthly rainfall values, originally in meters(m), are converted to millimeters (mm) for standardization. These values are then aggregated by summing them to produce total rainfall amounts for each month of each district across multiple years.

The graph 2.10 illustrates the rainfall data of Tanzania, the left side shows the monthly average rainfall for the years 2011, 2015, 2020, 2022, and 2023. The data shows a distinct seasonal pattern, with two main peaks of rainfall occurring, indicating a bimodal rainfall distribution. The first peak generally occurs between March and May, aligning with the long rainy season (Masika). The second peak occurs from October to December, aligning with the short rainy season (Vuli). The dry season from June to August consistently shows minimal rainfall across all years. Additionally, the right side of the graph presents the overall distribution of the rainfall data across all years. The histogram and KDE plot reveal that the rainfall data is right-skewed, indicating that most measurements are relatively low, with occasional higher rainfall values.

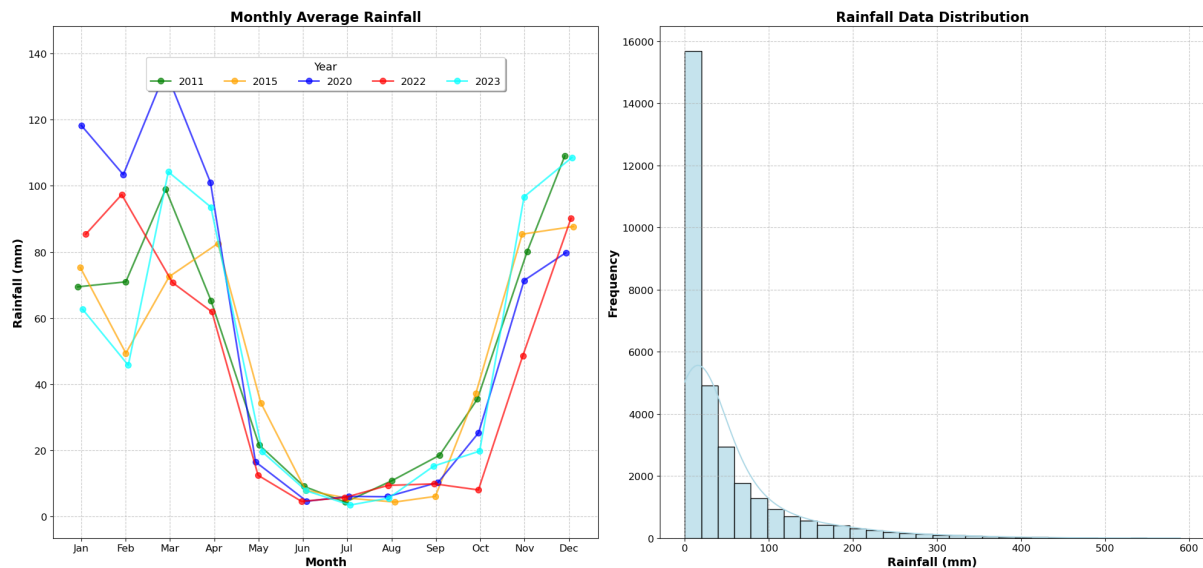


Figure 2.10: Tanzania Rainfall Data

The graph 2.11 illustrates the rainfall data for Rwanda, the left side shows the monthly average rainfall for the years 2006, 2012, 2015, 2018, and 2021. Similar to Tanzania, the data for Rwanda shows a distinct bimodal rainfall distribution. The first peak generally occurs between March and May, corresponding to the long rainy season and the second peak occurs from September to November, aligning with the short rainy season in Rwanda. The dry season from June to August consistently shows minimal rainfall across all years. Additionally, the right side of the graph presents the overall distribution of the rainfall data across all years. The histogram and KDE plot reveal a right-skewed distribution, indicating that most measurements are relatively low, with occasional high rainfall values extending the tail.

In general, both Rwanda and Tanzania exhibit a bimodal rainfall distribution due to the movement of the Intertropical Convergence Zone (ITCZ), resulting in two distinct rainy seasons each year. Tanzania generally receives more annual rainfall compared to Rwanda, with significant

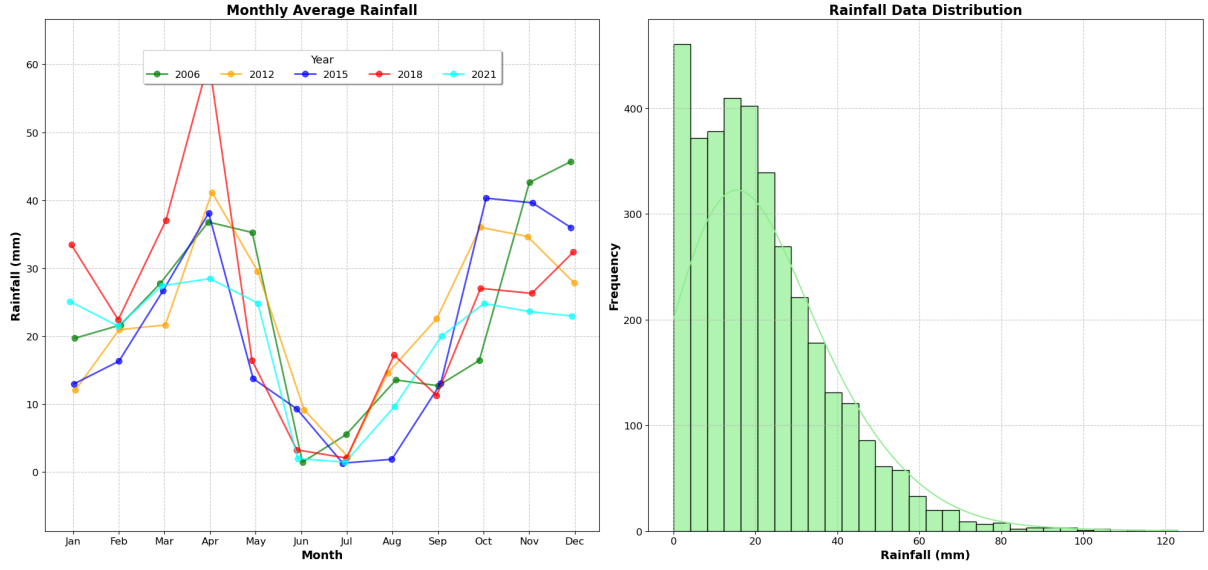


Figure 2.11: Rwanda Rainfall Data

peaks and extended rainy periods. Additionally, the overall distribution of the rainfall data in both countries is right-skewed, with most measurements being relatively low and occasional high rainfall events extending the tail. This analysis shows the importance of normalizing this data before training in the model to make sure that it provides accurate predictions.

Maximum and Minimum Temperature: The Minimum and Maximum Temperature data was collected from the ERA5 dataset [36] using similar methods as the rainfall data collection. This is 2m temperature data, measured at a height of two meters above the ground, which serves as a standard reference for meteorological observations and accurately reflects the surface air temperature experienced by crops and vegetation. The temperature values, initially measured in Kelvin, are converted to Celsius for standardization and easier interpretation. For each district, the minimum and maximum monthly temperatures are extracted and aggregated to capture the lowest and highest temperatures each month, resulting in a temporal resolution of one month per year.

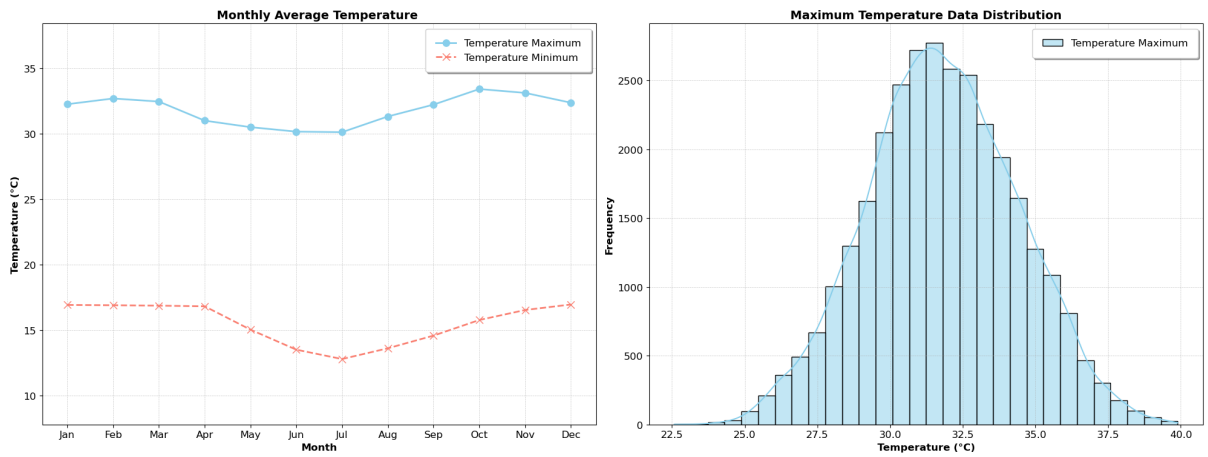


Figure 2.12: Tanzania Maximum and Minimum Monthly Temperature

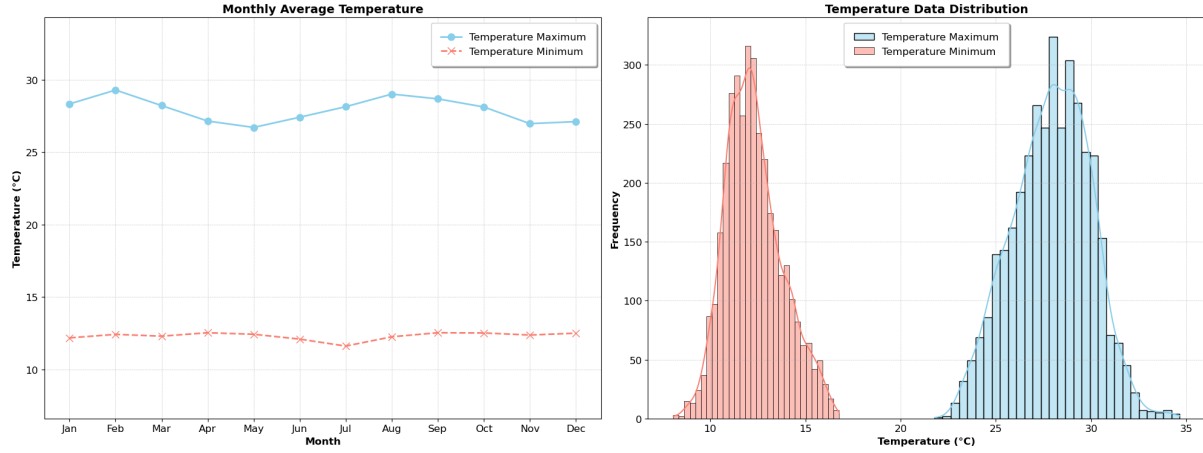


Figure 2.13: Rwanda Maximum and Minimum Monthly Temperature

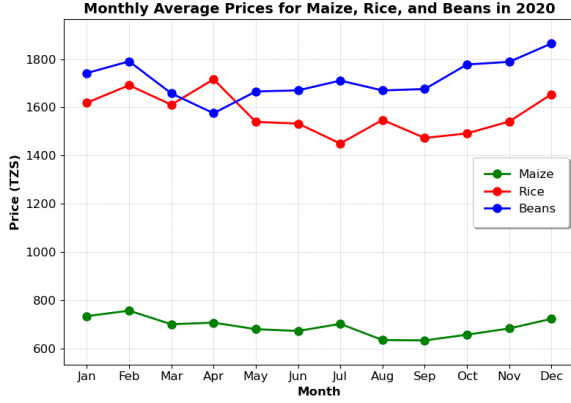
Figure 2.12 and 2.13 represent the temperature data for Tanzania and Rwanda respectively. Tanzania generally experiences higher temperatures, with the average maximum values ranging from approximately 30°C to 35°C and minimum values from 15°C to 20°C. In contrast, Rwanda has cooler temperatures, with maximum values ranging from 25°C to 30°C and minimum values from 10°C to 15°C. Both Tanzania and Rwanda exhibit a right-skewed distribution for their maximum temperature data, with Tanzania having a higher overall range of temperatures.

Food Price Data: The Food price data is collected from the World Food Programme (WFP) and is available at the Humanitarian Data Exchange for both Tanzania[38] and Rwanda [39]. This data, provided in CSV format, includes information on the prices of various food items such as beans, maize, and rice at the district level. However, some years have missing data, and certain districts lack complete records. To address these gaps, spatial interpolation using the K-nearest neighbors (KNN) algorithm is applied, with a distance parameter set to 5 kilometers. This method estimates missing values by considering the prices in neighboring districts within the specified distance. This approach ensures a more comprehensive and continuous dataset, which is crucial for accurate analysis and modeling. Using the food price data in our model helps predict food security indicators by providing insights into the economic accessibility of essential food items. Understanding price trends and variability is essential for assessing the affordability of food, which is a critical factor in predicting food security in different regions.

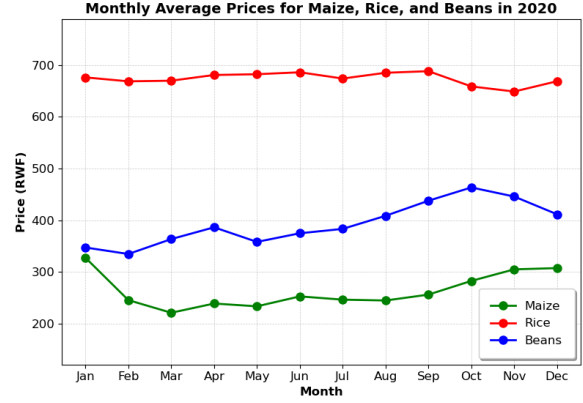
Figure 2.14(a) and Figure 2.14(b) represent the trends of prices of maize, beans and rice for Tanzania and Rwanda respectively. These trends show the average price of these food items on a country basis. In both countries, rice is the most expensive staple food. In Tanzania, rice prices range from 1400 to 1700 Tanzania Shillings(TZS) per 1 KG, while in Rwanda, they fluctuate between 680 and 720 Rwandan Franc (RWF). Beans are the second most expensive in both countries, with prices in Tanzania ranging from 1600 to 1800 TZS, and in Rwanda from 300 to 450 RWF. Maize remains the most affordable staple in both nations, with prices in Tanzania ranging from 600 to 800 TZS and in Rwanda from 200 to 300 RWF.

2.4.2 Conjunctural Data

Conjunctural data includes various indicators that are crucial for understanding the broader context of food security at the district level with a temporal resolution of one year and one value



(a) Tanzania Food Prices Trends in 2020

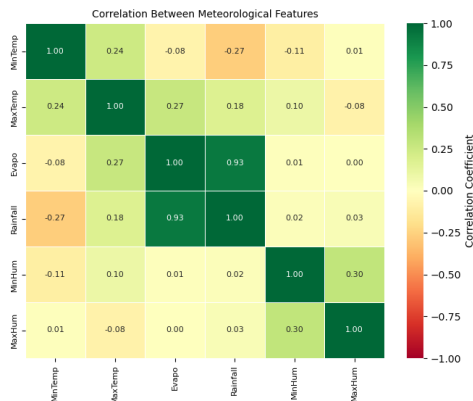


(b) Rwanda Food Prices Trends in 2020

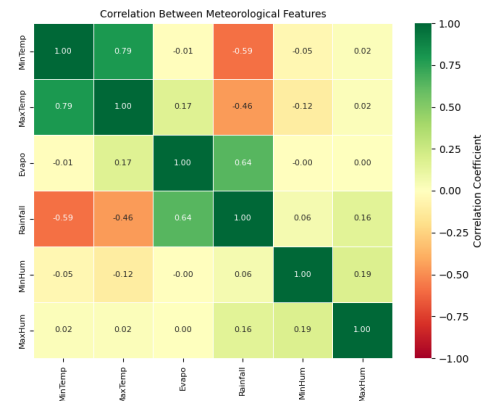
Figure 2.14: Food Prices Trends

per district. These datasets encompass meteorological, economic, and demographic variables, each providing a unique perspective on the factors influencing food security. The following sections will explore the details of each dataset, explaining the methods used for their collection and preprocessing to ensure accuracy and reliability for model training.

Meteorological Data: The meteorological data, including precipitation, evaporation, relative humidity, and temperature. These data are collected from the ERA5 dataset and extracted at the district level using the same method of clipping from the NetCDF file of the country data. Precipitation and evaporation data, initially measured in meters, were converted to millimeters and scaled by their respective sums for normalization. Temperature data, measured at two meters above the ground, were converted from Kelvin to Celsius, with minimum and maximum annual values aggregated to capture the lowest and highest temperatures each year. Relative humidity data were recorded in percentage form, directly reflecting atmospheric moisture content. This comprehensive data processing approach provides consistent and reliable measurements across different districts, with a temporal resolution of one year, ensuring each district has a single, annual value for each parameter.



(a) Tanzania Meteorological Data Correlation



(b) Rwanda Meteorological Data Correlation

Figure 2.15: Meteorological Data Correlation

To further investigate the relationships between these meteorological variables, we analyzed

the correlation matrix using the Pearson method to calculate the correlation coefficient. The Pearson correlation coefficient (r) measures the strength and direction of the linear relationship between two variables, ranging from -1 to 1. A value of 1 indicates a perfect positive relationship. The heatmap in the Figure 2.15(a) and 2.15(b) visually represents these coefficients, providing insight into the interdependencies among these variables. The correlation matrix for Tanzania shows a strong positive correlation between evaporation (Evapo) and rainfall (Rainfall) with a coefficient of 0.93 while in Rwanda the coefficient is 0.64 between these variables. This indicates that higher rainfall is associated with higher evaporation rates. Additionally, maximum temperature (MaxTemp) shows a moderate positive correlation with evaporation (Evapo) at 0.27 in Tanzania and 0.17 in Rwanda, suggesting that higher temperatures contribute to increased evaporation. On the other hand, minimum temperature (MinTemp) and rainfall (Rainfall) exhibit a moderate negative correlation of -0.27 in Tanzania and a stronger negative correlation of -0.59 in Rwanda, implying that lower minimum temperatures are often associated with higher rainfall. Relative humidity, represented by minimum humidity (MinHum) and maximum humidity (MaxHum), shows relatively weaker correlations with other variables, although there is a modest positive correlation of 0.30 in Tanzania and 0.19 between minimum and maximum humidity.

Population Data: The population data were collected from the Copernicus Emergency Management Service provided by the GHSL - Global Human Settlement Layer [40]. These data were gathered by downloading the tiles of TIFF files covering the areas near our study regions. These data have a resolution of 100m and consist of population estimates from the years 2000 to 2025, with intervals of five years (i.e., 2000, 2005, 2010, etc.). We used the country shapefiles to extract data aligning with the national boundaries of both Rwanda and Tanzania. To estimate population data for years not directly available (e.g., 2001, 2002, 2003, and 2004), we performed linear interpolation. This process involved reading existing TIFF files for known years, interpolating the data for the missing years, and generating new TIFF files. This method ensured accurate and reliable population estimates based on existing data, providing a continuous and consistent dataset for our analysis. After obtaining the TIFF files for each country, we used the clipping method to extract the population data for each district and subsequently computed various metrics, including population density, Gini coefficient, and spatial autocorrelation at 2 km and 5 km scales.

Population Density : is a metric that indicates the number of people living per unit area, helping to identify densely populated. The Population Density (PD) is calculated as follows:

$$PD = \frac{P}{A} \quad (2.3)$$

where P is the total population of the district and A is the area of the district.

Gini Coefficient : The Gini coefficient (G) measures the inequality of population distribution within a district. It ranges from 0 to 1, where $G = 0$ indicates perfect equality (each pixel/region in a district has the same population), $0 < G < 1$ indicates various levels of inequality in population distribution, and $G = 1$ indicates maximum inequality (all the population is concentrated in a single pixel). It is calculated as follows:

$$G = \frac{\sum_{i=1}^n (2i - n - 1)x_i}{n \sum_{i=1}^n x_i} \quad (2.4)$$

where n is the number of population values (pixels) within the district and x_i is the population

value of the i -th pixel, sorted in ascending order.

Spatial Autocorrelation : Spatial Autocorrelation measures the degree to which population values are correlated with themselves over space. It helps in understanding the spatial patterns and clustering of populations within districts. We calculated spatial autocorrelation at two different scales: 2 km and 5 km.

- **Autocorrelation at 2 km :** Provides insight into local clustering effects, showing how population values in one location are similar to those within a 2 km radius. Higher values indicate strong local clustering, where similar population densities are found close to each other.
- **Autocorrelation at 5 km :** Captures broader spatial patterns, indicating how population values correlate over a wider area. Higher values suggest broader regional clustering.

Economic Data: The economic data were downloaded from the World Bank database and include several key indicators which are directly downloaded separately as a CSV. For both countries (Rwanda and Tanzania), the values of all of these indicators for the year 2023 were missing and therefore we use linear interpolation to predict the missing values to obtain a complete dataset. These indicators are critical for understanding the economic context of the countries and include the following:

- GDP growth (annual %)^[41]: The annual percentage growth rate of GDP is calculated at market prices using constant local currency values. The aggregates are expressed in U.S. dollars, based on constant 2015 prices. GDP represents the total gross value added by all resident producers in the economy, including any product taxes and excluding subsidies that are not included in the product values
- GDP per Capita (annual %)^[42]: The annual percentage growth rate of GDP per capita, calculated in constant local currency, represents the GDP divided by the midyear population reflecting the average economic output per person.
- Trade (% of GDP)^[43]: The sum of exports and imports of goods and services measured as a share of GDP.
- Growth National Expenditure(% of GDP) ^[44]: is the total of household consumption, government spending, and capital investments. The annual growth rate of this expenditure reflects changes in economic spending patterns.
- Household Final Consumption Expenditure (% of GDP) ^[45]: represents the market value of all goods and services bought by households, including durable goods like cars, washing machines, and computers.
- Inflation, Consumer Prices (annual %)^[46]: represent the annual percentage change in the cost to the average consumer of acquiring a basket of goods and services, which may be fixed or adjusted at specified intervals, such as annually.
- Population Growth (annual %)^[47]: The annual growth rate of the population. The population is defined by the de facto method, which includes all residents regardless of their legal status or citizenship
- Unemployment(% of total labor force)^[48]: The percentage of the total labor force that is unemployed but actively seeking employment and willing to work.

- FDI Outflows (% of GDP) ^[49]: Foreign direct investment (FDI) refers to net investment outflows aimed at acquiring a lasting management interest in an enterprise operating in another economy. It includes equity capital, reinvested earnings, and other capital. A direct investment relationship is established when an investor owns 10% or more of the voting stock in the enterprise. These net outflows are typically measured relative to the GDP of the reporting economy
- FDI Inflows (% of GDP) ^[50]: Foreign direct investment (FDI) represents net inflows aimed at acquiring a lasting management interest (10% or more of voting stock) in enterprises in another economy. It includes equity capital, reinvested earnings, and both long-term and short-term capital as recorded in the balance of payments. The series shows net inflows (new investments minus disinvestments) from foreign investors into the reporting economy, relative to GDP.
- Merchandise Trade (% of GDP) ^[51]: The sum of exports and imports of goods, excluding services divided by the value of GDP.
- Military Expenditure (% of GDP) ^[52]: The financial resources a nation dedicates to raising and maintaining armed forces or other essential military resources.

NDVI Data: The NDVI data were collected from the MODIS Data Product. Specifically, the data were obtained from the MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061 dataset [53] for Rwanda however due to computational capacity limitations the L3 Global 500m SIN Grid V061 dataset[54] were collected for Tanzania. These data were accessed through API from the Application for Extracting and Exploring Analysis Ready Samples (AppEEARS) platform, provided by NASA’s EOSDIS Land Processes Distributed Active Archive Center (LP DAAC) [55]. NDVI is a critical indicator for assessing vegetation health and biomass, in the context of food security, high NDVI values correlate with healthy crops and abundant vegetation, which are essential for agricultural productivity. NDVI values range from -1 to 1, with values closer to 1 indicating healthy, dense vegetation and values closer to -1 indicating little or poor vegetation. The NDVI is computed using the following formula:

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)} \quad (2.5)$$

where *NIR* is the reflectance in the near-infrared spectrum and *RED* is the reflectance in the red spectrum.

The graph 2.16 represent the NDVI data for Tanzania in 2021 which reveals the significant seasonal variations in vegetation health and density, closely linked to the Tanzania’s rainfall patterns. During the wet seasons from November to May, NDVI values are higher, indicating denser and healthier vegetation due to increased rainfall. The peak values observed in March and April correspond with the long rains (Masika), while another increase in November and December aligns with the short rains (Vuli).

2.4.3 Spatial Data

This category includes data with spatial information, invariant per year, with one value per district. It includes data on hospitals, schools, and violent events soil quality, and elevation data.

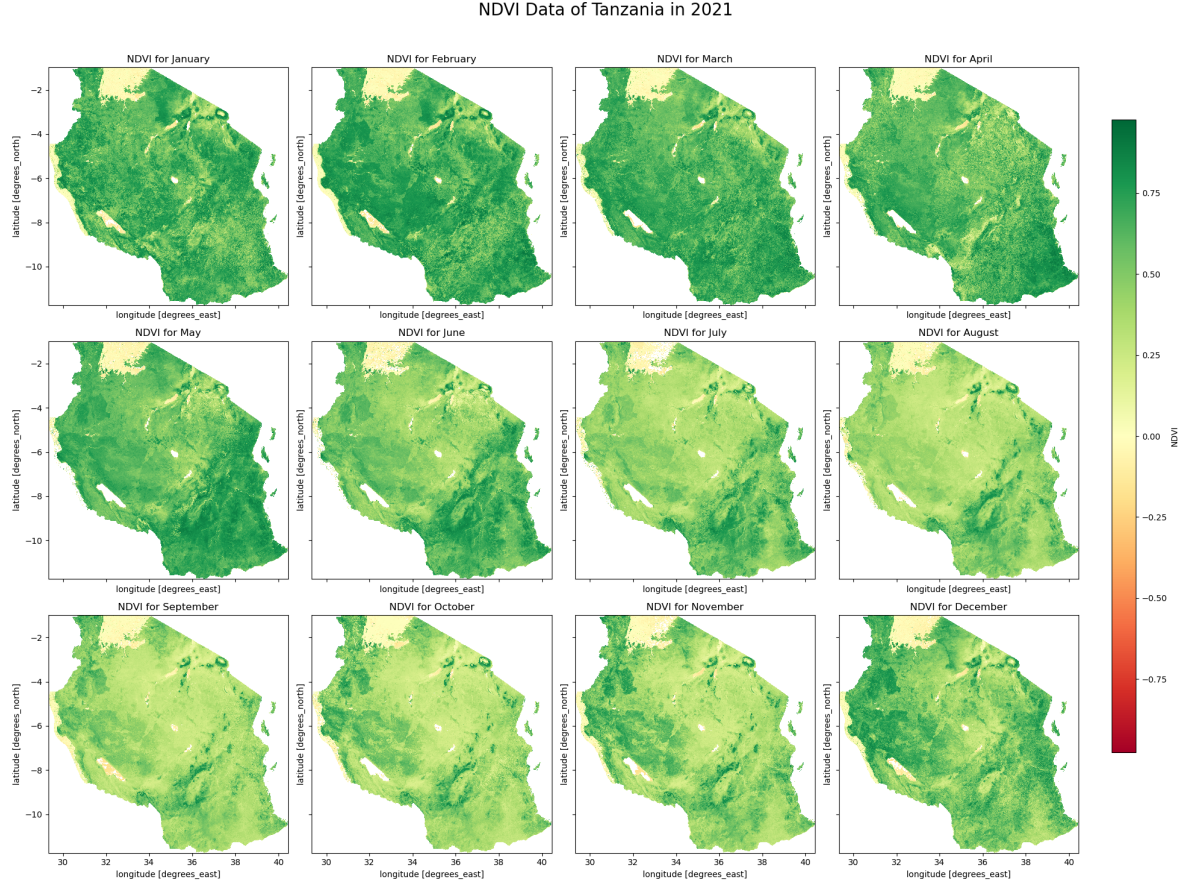


Figure 2.16: NDVI Data of Tanzania in 2021

Number of Hospital and Schools Per District: The spatial data for hospitals and schools was collected from OpenStreetMap (OSM) [56]. These data are in the form of points that represent amenities such as schools, colleges, hospitals, and clinics. To determine the number of each type of amenity within a district, we performed queries in OSM using API. This process involved using the geographical boundaries of the district polygons to extract relevant data points and then we counted the number of each amenity type, such as hospitals, clinics, schools, colleges, and universities, that fell within these district boundaries.

Waterways Data: Waterways data are represented in the form of line vectors. They include data such as rivers, streams, canals, lakes, and others. These data are also obtained from OSM using the same method as applied while extracting hospital and school data. To obtain the spatial distribution of waterways within a district we compute the waterways density per square kilometer by taking the total number of waterways available in each district divided by the area of the district (in kilometer square). Mathematically, The density of waterways (DW) in each district is calculated as follows:

$$Nw_d = \sum_{i=1}^I w_i \quad (2.6)$$

where Nw_d is the total number of waterways intersecting the district boundary, w_i is a binary indicator function that equals 1 if the i -th waterway intersects the district d , and I is the total

number of waterways.

The area (A_d) of each district in square kilometers is given by:

$$A_d = \frac{A_d^{\text{m}^2}}{1,000,000} \quad (2.7)$$

where $A_d^{\text{m}^2}$ is the area of district d in square meters. The density of waterways (DW) per square kilometer is then calculated by:

$$DW = \frac{Nw_d}{A_d} \quad (2.8)$$

Violent Events Data: Violent events data are collected from the Armed Conflict Location and Event Data Project (ACLED)[57], a reliable dataset that tracks conflict-related incidents globally. It provides detailed information on the occurrence, type, and location of various conflict events, including battles, protests, riots, and violence against civilians. By integrating this data we can gain valuable insights into how conflicts and violent events impact agricultural production and food distribution. To obtain the yearly number of violent events, we count the number of events recorded in each district. However, in both Rwanda and Tanzania, the number of these violent events appears to be very low, indicating that not many violent events are happening in these countries

Soil Quality Data: Soil quality data was collected from the Harmonized World Soil Database (HWSD)[58], developed by the Food and Agriculture Organization (FAO) and the International Institute for Applied Systems Analysis (IIASA). This database provides comprehensive global soil information, seven key soil qualities important for crop production have been derived from this database, however, in our study, we only focus on three qualities namely nutrient retention capacity, rooting conditions, and oxygen availability to roots. Nutrient retention capacity indicates the ability of a soil to retain its essential nutrients, which is vital for plant growth and the sustainability of agricultural practices. Rooting Conditions a metrics that describe the physical properties of soil that affect root growth, such as soil depth, texture, and structure. Oxygen availability to roots is a metric that asses the ability of soil to provide adequate oxygen to plant roots, necessary for respiration and overall plant health.

The data were initially downloaded as TIFF files with a spatial resolution of 1 kilometer(km), each representing global data for one of the three selected soil qualities. To analyze these data at the district level, we first used clipping to extract the relevant data for each country. Subsequently, we merged the three separate TIFF files into a single TIFF file containing three bands, with each band representing one of the selected soil quality metrics. This merged TIFF file was then used to extract the values for each district, allowing for a detailed assessment of soil quality at a district level.

Elevation Data: Elevation data is collected using API from the AppEEARS platform. This dataset provides a Digital Elevation Model (DEM) with a 90-meter resolution, derived from the Shuttle Radar Topography Mission (SRTM) conducted in the year 2000 [59]. The data were initially in NetCDF format, and the clipping method was used to extract the data for each district. For each district, we computed the maximum and variance of the elevation data. This data plays a significant role in food security as it influences climatic conditions. Areas with higher elevations may experience different climatic conditions compared to lower regions, affecting the types of crops that can be grown and their yield. It may also be the soil quality

and water availability which are critical factors for agricultural productivity.

2.4.4 High Spatial Resolution Data

This category comprises high-resolution spatial data, characterized by multiple values per district, and is presented in raster format (images). These raster images were designed for comprehensive spatial analysis and were processed using image processing techniques for data extraction and analysis. They are categorized as high spatial data as they provide high precision and detailed granularity representing spatial attributes across different districts. The data include the following:

Population Data: The population data used here are the same as those collected from GHSL [40], as explained in the Conjunctural Data section, where they were processed to compute metrics such as population density, Gini coefficient, and spatial autocorrelation. However, in this category, we utilize the original raster format with a spatial resolution of 100m. Each raster image maintains the visual and geographic precision of demographic data, facilitating detailed examinations of population distribution and growth patterns across various districts. Understanding this distribution is crucial for assessing its impact on resource allocation and the human dimension of food security. Moreover, it allows for integration with other spatial datasets, such as Land Use Land Cover (LULC) data, enabling comprehensive spatial analyses that consider multiple variables simultaneously.

Land Use Land Cover Data(LULC): The land use and land cover data were collected from Esri’s Sentinel-2 dataset [60], with a spatial resolution of 10 meters. The data were manually downloaded as tiles in the form of TIFF files and then merged to extract the LULC for our study areas. We use Sentinel-2 LULC data because it provides high spatial resolution, enabling us to capture detailed patterns of land use and land cover changes that are essential for accurate geographic representation. This dataset consists of different classes, as illustrated in Figure 2.17, but for our task, we focus on three classes: crops, forests, and built areas. These classes are crucial for understanding their significant influence on environmental and socioeconomic factors such as agricultural productivity, biodiversity conservation, and urban development. Their impact on agricultural productivity directly relates to food security, as crop areas determine the potential for food production, forest areas affect ecosystem services and biodiversity, and built areas indicate the extent of urbanization and its effects on agricultural land availability and food distribution networks.

We processed the LULC data by extracting specific class values corresponding to crops, forests, and built areas to obtain these three classes as shown in Figure 2.18. This involved converting the original raster data into binary rasters where the presence of a particular class is marked as 1 and its absence as 0. We then resampled these binary rasters to a 100-meter resolution using Average as a resampling method to create fractional coverage rasters of float values.

After resampling, we obtained three raster datasets for forests, crops, and built areas. To integrate these with population data, we ensured all datasets had consistent spatial dimensions and coordinate reference systems (CRS). We rescaled the population raster data, downsampled it using the nearest neighbor method, and reprojected it into the same CRS as the LULC rasters. This alignment is crucial for accurate spatial analysis and overlay operations within district boundaries.

RWANDA LAND USE LAND COVER 2020

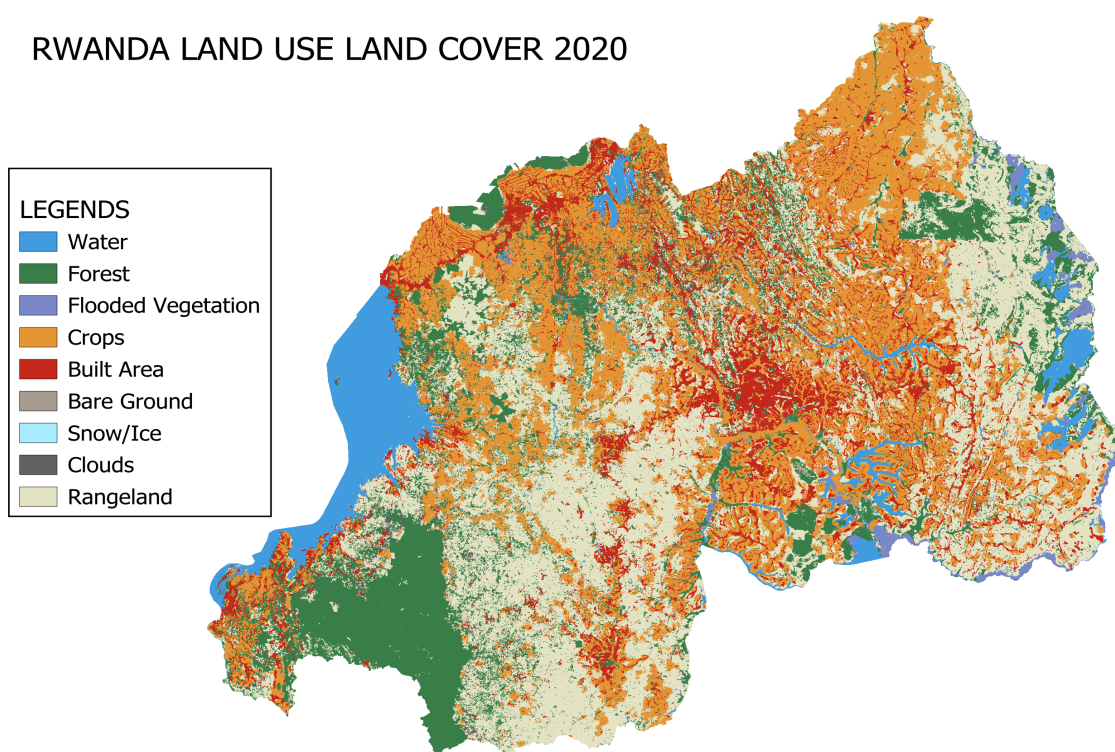


Figure 2.17: Raster Of Land Use Land Cover Data of Rwanda

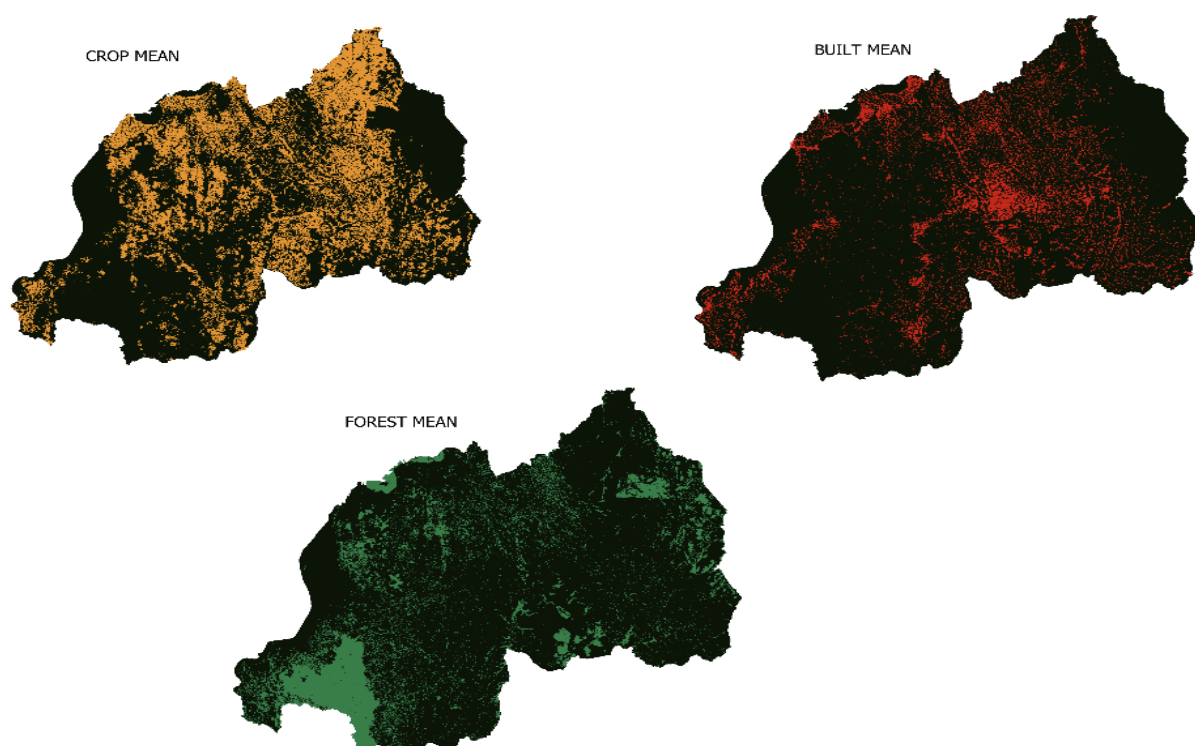


Figure 2.18: Raster Mean After Resampling To 100m

The table 2.5 summarize the key characteristics of all feature datasets used, including their resolution, source, and scaling-up method.

Data	Spatial Resolution	Temporal Resolution	Data Source	Scaling Method Up
Time series data (several values per year, one value per district)				
Smoothed brightness temperature(K) $\rightarrow$ ($^{\circ}$ C)	0.07 $^{\circ}$	1 month	NOAA	Maximum
Rainfall (m) $\rightarrow$ (mm)	0.25 $^{\circ}$	1 month	ERA5	Sum
Average minimum and maximum temperatures(K) $\rightarrow$ ($^{\circ}$ C)	0.25 $^{\circ}$	1 month	ERA5	Mean
Beans, Maize and Rice price	NA	1 month	WFP	K-nearest neighbor interpolation
Conjunctural data (one value per year, one value per district)				
Meteorological data	0.25 $^{\circ}$	1 year	ERA5	Sum, Minimum, Maximum
Population density	100m	1 year	GHSL	Spatial autocorrelation 2km and 5km, Gini Coefficient
Economic data	Country	1 year	World Bank	Country value
Normalized difference vegetation index	250m (Rwanda) 500m (Tanzania)	1 month	MODIS	Mean
Spatial data (one value per district)				
Hospitals, schools	Point vectors	2024	Open Street Map	Count
Violent events	Point vectors	2018	ACLED	Count
Soil quality	1km	2008	FAO	Mean
Waterways	Line vectors	2024	Open Street Map	Count, length
Elevation data	90m	2000	NASA	Maximum, Variance
High spatial resolution data (several values per district)				
Population density	100m	1 year	GHSL	CNN
Land cover (crops, forests, building areas)	10m $\rightarrow$ 100m	2020	ESRI	CNN

Table 2.5: Summary of the Feature Data with their Category

CHAPTER 3

Methodology

3.1 General Framework

The general framework of our study comprises three different branches, as illustrated in Figure 3.1. The time series branch is modeled using a Long Short-Term Memory (LSTM) network, the high spatial resolution data branch is handled by a Convolutional Neural Network (CNN), and the conjunctural and spatial data branch is addressed using a Random Forest (RF) model. This framework is designed to offer dual prediction capabilities, enabling both regression and classification tasks. The implementation is carried out using Python, with PyTorch serving as the primary library for the deep learning models (LSTM and CNN), while the RF model is implemented using the scikit-learn library, which is well-suited for traditional machine learning tasks.

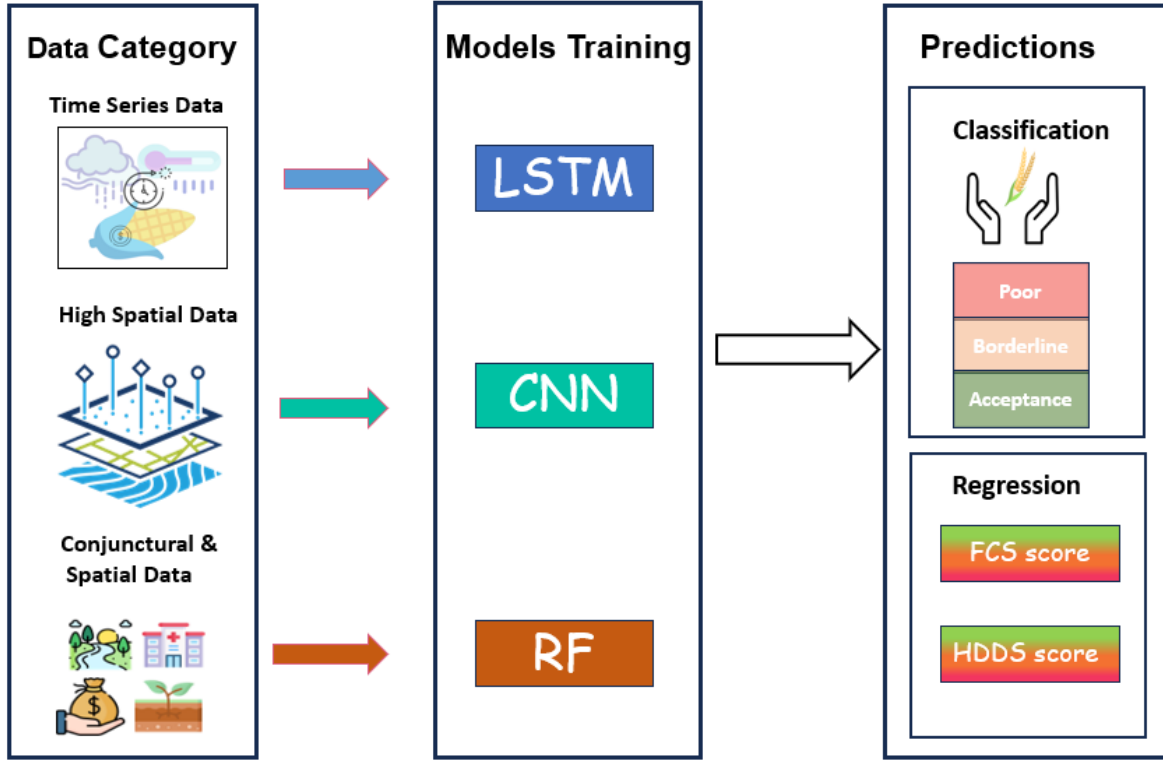


Figure 3.1: The General Framework

3.2 Feature Engineering

This section details the various preprocessing steps undertaken to prepare the data for developing predictive models. As explained in the previous chapter, feature data are categorized

into four different groups: time series, conjunctural, spatial, and high spatial. Given the diverse sources, formats, and thematic content of these datasets, further processing is essential to ensure accurate predictions. Therefore, we applied various preprocessing strategies to prepare the data for model training to ensure that our models effectively enhance the accuracy of our predictions but also directly contribute to answering our second research question *RQ2: Which specific data adaptations are needed to apply within the FSPHD framework for other African countries with different climatic and socio-economic conditions?*. These strategies include:

3.2.1 Normalization and Scaling

Normalization and scaling are crucial to ensure that all features contribute equally to the model training process. Different scaling methods were applied to address the diverse statistical properties and distribution characteristics of the data. This step is essential to improve the model's performance by ensuring that no single feature dominates due to its scale. In the previous work by [1], all four data categories were normalized using Standard Scaler. However, this approach assumes that all features follow a normal distribution, which might not be appropriate for the diverse nature of the heterogeneous data involved in this study. Therefore for our study, we decided to explore the use of three different scaling methods; Standard, Min-Max, and Robust Scaler. Each scaler has its advantages and disadvantages, tailored to handle different data distribution patterns. For instance, while the Standard Scaler is more sensitive to outliers, the Robust Scaler can effectively manage scaling in the presence of outliers, and Min-Max can be used when it's crucial to maintain the relative scale of different features. The idea to use these different scalers is to determine which scaler is better suitable for each category of the data that could provide a better performance of the model. These scaler methods can be defined as follows:

- a) **Standard Scaler** Standard Scaler is a mean-based scaling method used to transform features by centering them around the mean and scaling them to have a standard deviation of one. The formula for scaling using Standard Scaler is:

$$X' = \frac{X - \mu}{\sigma}$$

where X' is the standardized value, X is the original value, μ is the mean of the feature, and σ is the standard deviation of the feature

- b) **Min-Max Scaler** Min-Max Scaler is a method used to transform features by scaling each feature to a given range, usually between 0 and 1. The formula for scaling using Min-Max Scaler is:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

where X' is the scaled value, X is the original value, $X_{\min}$ is the minimum value of the feature, and $X_{\max}$ is the maximum value of the feature.

- c) **Robust Scaler** Robust Scaler is a method used to transform features by removing the median and scaling them according to the interquartile range (IQR). The formula for scaling using Robust Scaler is:

$$X' = \frac{X - Q_2}{Q_3 - Q_1}$$

where X' is the scaled value, X is the original value, Q_2 is the median of the feature, Q_1 is the first quartile (25th percentile) of the feature and Q_3 is the third quartile (75th percentile) of the feature.

3.2.2 Time Window Selection

Selecting an appropriate time window for time-series data is crucial for accurately capturing seasonal and annual variations, particularly in our study where sowing and harvesting are significantly impacted by these variations. Therefore, we determined the time window based on the survey year and the rainy seasons specific to each country. Both Rwanda and Tanzania experience bimodal rainfall patterns, with long rains from March to May and short rains from October to December. For a survey conducted in year y , we selected the time window from March of the year $y-1$ to April of the year y . This window captures the entire agricultural cycle, integrating all relevant data variables such as temperature, rainfall, and food prices, ensuring the selected ML model has comprehensive temporal data to learn the impact of seasonal variations on food security indicators effectively.

3.3 Model Selection, Training, and Evaluation

This section provides details on the model selection, training, and evaluation processes used to predict the FCS and HDDS. This approach allows us to explore and optimize various ML and DL techniques within the FSPHD framework, helping to answer our third research question: *RQ3: Which aspects of machine learning and deep learning techniques can be optimized within the FSPHD framework to improve performance in predicting food security indicators?*. But also helps to answer the fourth research question *RQ4: How can the FSPHD framework be adapted to incorporate both regression and classification tasks effectively?*

3.3.1 Model Selection

Given the diverse nature of our datasets, we selected different models for each of the four data categories: Time Series, Conjunctural, Spatial, and High Spatial. We systematically tested various data preparation strategies, including three data-splitting techniques that we will explain in the next section combined with normalization methods, to evaluate and optimize model performance. Additionally, we conducted hyperparameter tuning to fine-tune the models, ensuring that each model was optimally configured for its specific data category.

Timeseries Data

For time-series data, we conducted experiments using LSTM networks and RF. These models were chosen because of their complementary strengths in addressing the unique challenges of time series data.

A Long Short-Term Memory (LSTM) network is a type of Recurrent Neural Network (RNN) designed to capture long-term dependencies and sequences in data. Its architecture utilizes memory cells with three gates: an input gate (controls what information is added to memory), a forget gate (controls what information is removed from memory), and an output gate to regulate information flow. This makes LSTMs effective for time series, machine translation, and speech recognition.[61]

On the other hand, Random Forest (RF) is an ensemble ML model comprising a collection of decision trees during training and outputting the average prediction of the individual trees. RF

can handle complex datasets and mitigate overfitting, making it a valuable model for various predictive tasks in machine learning. It is most commonly used for both regression and classification tasks. By using many decision trees, RF can model complex, non-linear relationships and make it less prone to overfitting than a single decision tree. This makes RF a good candidate for time-series data.

High Spatial Resolution Data

Convolution Neural Networks were employed to process and analyze high-resolution spatial data (i.e., LULC and Population Raster), which often contain complex spatial relationships that are critical for accurate predictions. CNNs were chosen because of their effectiveness in processing raster data, such as satellite imagery or other spatial datasets, due to their ability to capture spatial hierarchies and local patterns within the data. The effectiveness of CNNs for LULC has been demonstrated in various studies, such as the one by [62], which highlights their capability to accurately classify LULC and crop identification using hyperspectral remote sensing images. Also, the study by [1] demonstrates the use of CNNs for extracting features from population and LULC rasters, additionally the versatile architecture of CNN further supports their suitability for our application.

Conjunctural and Spatial Data

For the Conjunctural and Spatial data categories, we have chosen RF. The decision to select RF is based on its effective use for both regression and classification tasks, and interoperability. It is also well-suited for handling complex datasets with multiple features and interactions.

3.3.2 Data Splitting Techniques

The following are the three different data splitting techniques that were applied to split the data for model training and evaluation. The idea is to ensure that each technique serves a unique purpose and helps to determine the robustness and reliability of the framework.

- **Percentage-Based Split:** In this technique, the data is partitioned into training and testing sets based on a predefined ratio. We selected the data randomly where 85% of the data was used for training, while the remaining 15% was reserved for testing. This approach ensures that the model has sufficient data for learning while also preserving a portion for validation.
- **Temporal-Based Split :** This method involves dividing the data based on specific periods, ensuring that the training and testing sets represent different temporal segments. Specifically, we used the last year of the survey data as the testing set, with all preceding years used for training. For instance, if we have survey data from years $y_1, y_2, \dots, y_n$, we use y_n as the testing set and $y_1, y_2, \dots, y_{n-1}$ as the training set. This approach is essential for models predicting future outcomes based on historical data, as it helps evaluate the model's performance on unseen future data.
- **Spatio-temporal Split :** For this technique, we incorporated both spatial and temporal data splitting processes. For each year in the dataset, we selected 85% of the districts

for the training set and the remaining 15% for the testing set. For example, if we have districts (A, B, C, D, E, F) , districts (A, B, C, D) will be used for training, while the remaining districts $(E \text{ and } F)$ will be used for testing each year. This method ensures that the model is trained on a diverse set of geographical areas while also preserving a portion of the spatial data for testing. This approach evaluates the model's ability to generalize across different spatial regions.

3.3.3 Evaluation Metrics

Classification

This task involves predicting the food security group, which is categorical. It is a multi-class problem where we have defined three groups: Poor, Borderline, and Acceptance. These groups were established during the processing of ground truth data for both FCS and HDDS, based on the thresholds assigned for each country. The following metrics were used to evaluate the performance of our classification models:

- **Accuracy:** Measures the proportion of number of correct predictions by the total number of predictions

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** Indicates the proportion of true positive predictions out of all positive predictions made by the model.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):** Reflects the proportion of true positive predictions from all actual positive instances in the dataset.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1 Score:** A harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

Regression

This task involves predicting continuous outcomes. In our study, this includes predicting the FCS and HDDS. The regression problem requires the model to output a continuous value that can vary over a range, rather than fitting the data into discrete categories. To evaluate the performance of our regression models, we used two key metrics that provide insights into the accuracy and reliability of the predictions. The following metrics were used to assess our regression models:

- **Mean Squared Error (MSE)**: Measures the average squared difference between the predicted values and the actual values.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the number of data points.

- **R-squared (R^2)**: Indicates the proportion of the variance in the dependent variable that is predictable from the independent variables. It provides a measure of how well the model fits the data.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the actual values.

CHAPTER 4

Results and Discussion

4.1 Experimental Results

In this section, we present the experiments' results to evaluate the performance of the selected models of various data categories on the prediction of FCS and HDDS. The experiments were designed to assess the models' effectiveness in regression and classification tasks, focusing on identifying the most generalized scaling method applicable across all countries. The generalized scaling method was then applied to all the countries and evaluated the performance of the model using all three data-splitting techniques; percentage, temporal, and spatio-temporal. These results are organized into two main subsections regression results which are evaluated by R^2 metric and classification results which are evaluated by accuracy.

4.1.1 Regression Results

Time series Data

Table 4.1 presents results obtained with time-series data by using two models, the RF and LSTM model using a Min-Max scaler. Note that the results of the experiments included in the appendix D demonstrate that the Min-Max scaler was observed to be a more generalized scaling method for both models across all the countries, making it the preferred choice for benchmarking purposes.

Time Series Data by RF and LSTM with R^2					
Country	Indicator	Model	Percentage	Temporal	Spatio-Temporal
Rwanda	FCS	RF	0.346	0.351	0.286
		LSTM	0.282	0.409	0.036
	HDDS	RF	0.257	-0.17	0.602
		LSTM	0.647	0.321	0.518
Tanzania	FCS	RF	0.234	0.038	0.264
		LSTM	0.25	0.043	0.021
	HDDS	RF	0.121	-0.122	0.225
		LSTM	0.217	0.014	0.088
Burkina Faso	FCS	RF	0.233	-0.023	0.101
		LSTM	0.151	0.018	0.158
	HDDS	RF	0.177	0.077	0.184
		LSTM	0.225	0.194	0.231

Table 4.1: Comparative Regression Results of RF and LSTM Models for Timeseries Data in Rwanda, Tanzania, and Burkina Faso

The results demonstrate that both RF and LSTM models exhibit poor performance with variation depending on the data-splitting technique, indicator, and country. The highest R^2 value for FCS obtained was observed in Rwanda with a value of **0.351** and **0.409** using RF and LSTM respectively with a temporal split. The remaining countries got the R^2 score of less than **0.3** for FCS and HDDS across all the splits. Generally, when comparing the overall performance of RF and LSTM, the LSTM model performed slightly better across all the countries and indicators.

Additionally, Figure 4.1 demonstrates how the performance of the LSTM model in predicting FCS in Tanzania varies significantly across different data-splitting techniques. The results clearly show that the choice of data-splitting method can substantially affect the model's predictive power, with the Percentage split yielding the highest R^2 value of **0.25**, while the Temporal and Spatio-Temporal splits perform noticeably worse, with R^2 values of **0.043** and **0.021** respectively. These findings emphasize the importance of carefully selecting an appropriate data-splitting strategy to optimize model performance.

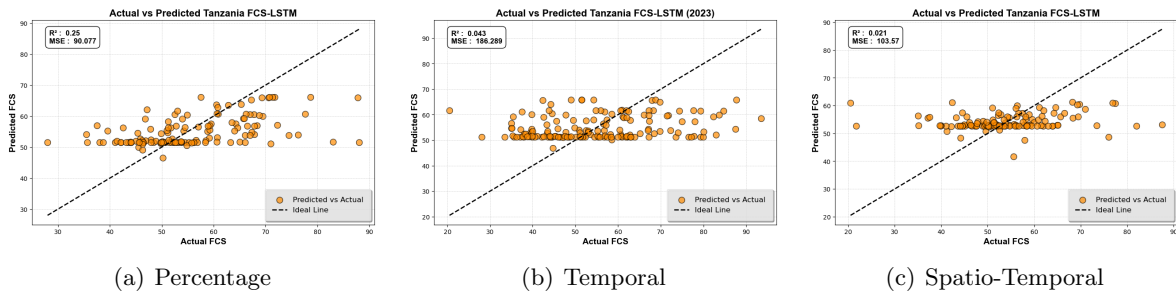


Figure 4.1: Comparison of Actual vs. Predicted FCS using Times Series Data by LSTM Model in Tanzania with different Data Splitting Techniques

High Spatial Resolution Data

The Table 4.2 illustrate the performance of CNN on high spatial data by using Robust Scaler across all the countries. The results demonstrate that the CNN achieved better performance by using temporal splits across all the countries and all the indicators. For example, for FCS the R^2 achieved is **0.662**, **0.655** and **0.766** for Rwanda, Tanzania and Burkina Faso respectively. Similarly for HDDS, the R^2 obtained is **0.728**, **0.632** and **0.78** for Rwanda, Tanzania and Burkina Faso respectively.

High Spatial Resolution Data by CNN with R^2				
Country	Indicator	Percentage	Temporal	Spatio-Temporal
Rwanda	FCS	0.512	0.662	0.432
	HDDS	0.623	0.728	0.227
Tanzania	FCS	0.415	0.655	0.083
	HDDS	0.435	0.632	0.203
Burkina Faso	FCS	0.614	0.766	0.434
	HDDS	0.682	0.78	0.598

Table 4.2: Comparative Regression Results of CNN Model for High Spatial Resolution Data in Rwanda, Tanzania and, Burkina Faso

Conjunctural and Spatial Data

The table 4.3 present the experiment results of conjunctural and spatial data by using RF model. The Robust scaler was selected for this category as it proved to be the most generalized scaling method across all countries. The choice of this scaler is explained in the preliminary results(before hyperparameter tuning) present in the appendix E.

The results demonstrate that the percentage split is more generalized across all countries, providing consistently better performance. For example, using the percentage split, the R^2 values for FCS are **0.59**, **0.346** and **0.386** for Rwanda, Tanzania, and Burkina Faso respectively. Although there are slight variations, such as Rwanda achieving a slightly higher R^2 of **0.6** with the temporal split and Tanzania reaching **0.357** with the spatio-temporal split, these differences are marginal.

Conjunctural and Spatial Data by RF with R^2				
Country	Indicator	Percentage	Temporal	Spatio-Temporal
Rwanda	FCS	0.59	0.6	0.441
	HDDS	0.809	0.554	0.629
Tanzania	FCS	0.346	-0.012	0.357
	HDDS	0.121	-0.104	0.103
Burkina Faso	FCS	0.386	0.268	0.096
	HDDS	0.299	0.289	0.244

Table 4.3: Comparative Regression Results of RF Model for Conjunctural and Spatial Data in Rwanda, Tanzania, Burkina Faso

Figure 4.2 illustrates the performance of RF on the prediction of HDDS for Rwanda 4.2(a), Tanzania 4.2(b) and Burkina Faso 4.2(c) by using Percentage splits. The graph shows the R^2 of HDDS is **0.809**, **0.121** and **0.299** with corresponding MSE values of **0.179**, **0.659** and **0.904** for Rwanda, Tanzania and Burkina Faso respectively. These results suggest that the model performs exceptionally well in Rwanda, achieving a high R^2 and low MSE, indicating strong predictive accuracy. However, the model's performance in Tanzania is notably weaker, as reflected by the low R^2 and higher MSE, indicating less reliable predictions. In Burkina Faso, the model's performance is moderate, compared to the results observed for Rwanda and Tanzania.

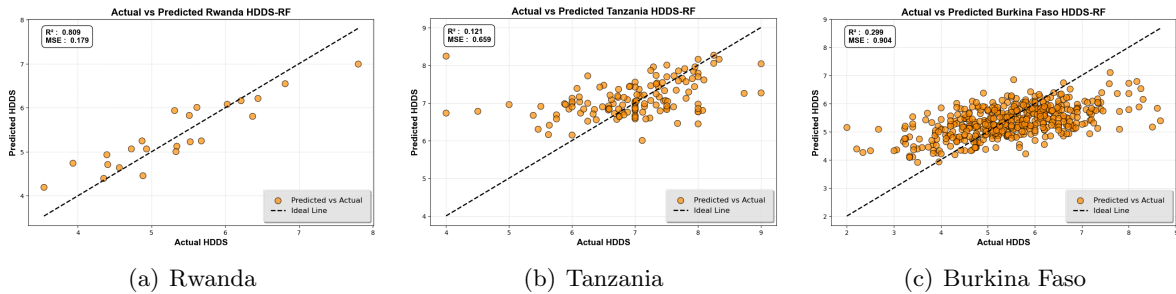


Figure 4.2: Comparison of Actual vs. Predicted HDDS using Conjunctural and Spatial Data by RF Model for Rwanda, Tanzania, and Burkina Faso with Percentage Split

4.1.2 Classification Results

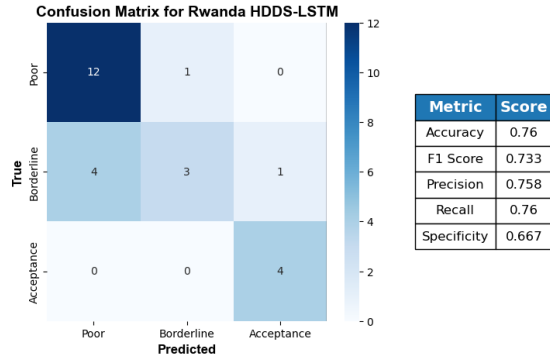
Time series Data

Table 4.4 presents the classification results of time series data using a Min-Max scaler with RF and LSTM models. The results demonstrate that LSTM generally outperformed RF across all countries and indicators, except for the HDDS indicator in Tanzania, where RF shows superior performance. For instance, in Rwanda, LSTM achieved the highest accuracy for FCS at **0.84** and for HDDS at **0.76** using the spatio-temporal split. Additionally, the results highlight the variability of model performance based on the country, data-splitting techniques, and indicators. For example, in Burkina Faso, LSTM achieved an accuracy of **0.516** for FCS using the temporal split and **0.523** for HDDS using the spatio-temporal split.

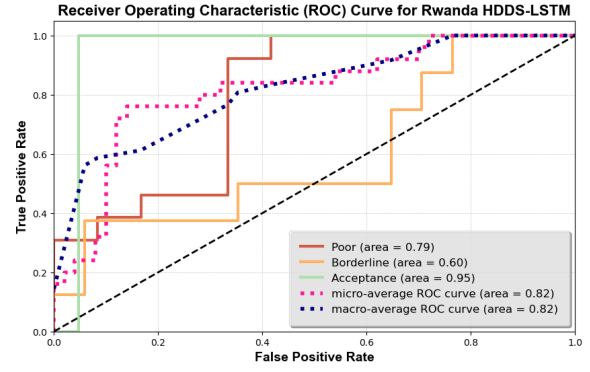
Time Series Data by RF and LSTM with Accuracy Metric					
Country	Indicator	Model	Percentage	Temporal	Spatio-Temporal
Rwanda	FCS	RF	0.609	0.517	0.6
		LSTM	0.552	0.69	0.84
	HDDS	RF	0.609	0.586	0.56
		LSTM	0.609	0.586	0.76
Tanzania	FCS	RF	0.511	0.468	0.444
		LSTM	0.541	0.513	0.559
	HDDS	RF	0.481	0.411	0.582
		LSTM	0.541	0.437	0.548
Burkina Faso	FCS	RF	0.452	0.418	0.43
		LSTM	0.498	0.516	0.515
	HDDS	RF	0.459	0.491	0.463
		LSTM	0.493	0.519	0.523

Table 4.4: Comparative Classification Results of RF and LSTM Models for Time series Data in Rwanda, Tanzania and Burkina Faso

Figures 4.3, 4.4 and 4.5 illustrate the performance of the LSTM model in classifying the HDDS for Rwanda, Tanzania and Burkina Faso, respectively, using the spatio-temporal split. Each figure presents a confusion matrix alongside corresponding performance metrics and a Receiver Operating Characteristic (ROC) curve, providing a comprehensive overview of the model's classification accuracy and ability to distinguish between different HDDS categories. The graphs demonstrate that the LSTM model achieved a robust performance in classifying the **Poor** and **Acceptance** HDDS categories across all the countries, as evidenced by the confusion matrices and high AUC values of each country. For example, Rwanda had AUC values of **0.79** for **Poor** and **0.95** for **Acceptance**. Similarly, Tanzania had AUC value of **0.72** for both **Poor** and **Acceptance** and Burkina Faso had **0.77** and **0.71** for **Poor** and **Acceptance** respectively. However, the model consistently struggles with the **Borderline** category, leading to a significant number of misclassifications, as reflected by lower AUC values of **0.60**, **0.50** and **0.60** for Rwanda, Tanzania and Burkina Faso respectively. In all three countries, **Borderline** instances are often incorrectly predicted as **Poor** or **Acceptance** resulting in lower performance metrics for this class.

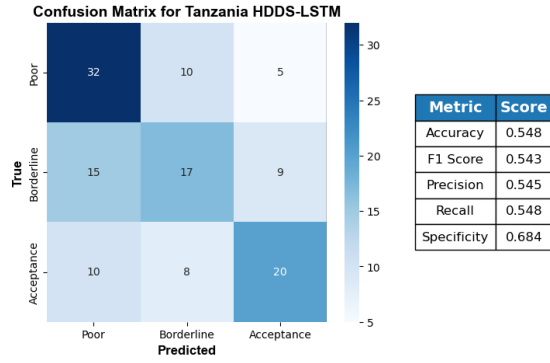


(a) Confusion Matrix and Performance Metrics

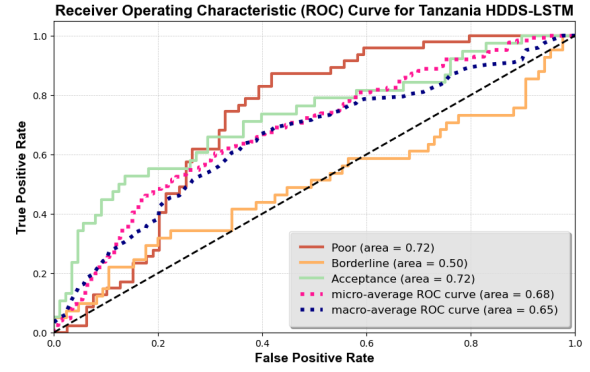


(b) ROC Curve

Figure 4.3: LSTM performance of HDDS for Rwanda with Spatio-Temporal using Time series Data

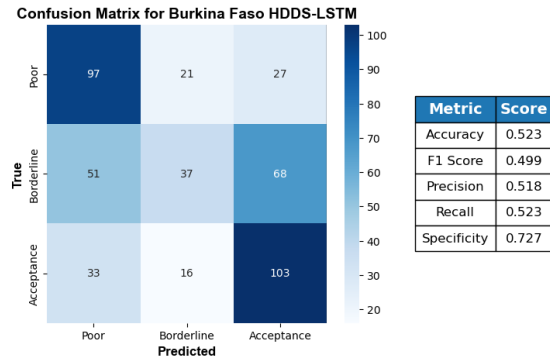


(a) Confusion Matrix and Performance Metrics

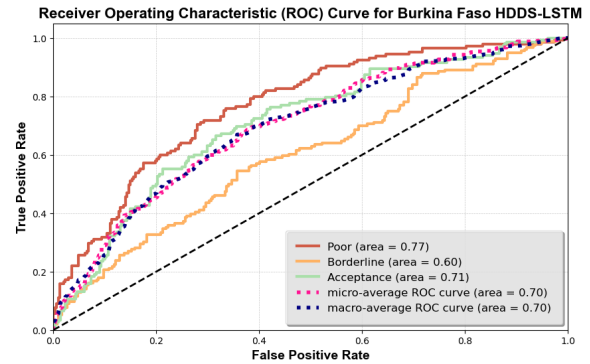


(b) ROC Curve

Figure 4.4: LSTM performance of HDDS for Tanzania with Spatio-Temporal using Time Series Data



(a) Confusion Matrix and Performance Metrics



(b) ROC Curve

Figure 4.5: LSTM performance of HDDS for Burkina Faso with Spatio-Temporal using Time series Data

High Spatial Resolution Data

Table 4.5 presents the classification results of the High Spatial Resolution Data using a CNN with a Robust Scaler. The results indicate that the CNN generally performed best with temporal splits across all countries, with a notable exception in Rwanda, where the model achieved perfect prediction with the spatio-temporal split. Using temporal splits, the model achieved accuracies of **0.966**, **0.822**, and **0.923** for FCS, and **0.966**, **0.808**, and **0.923** for HDDS in Rwanda, Tanzania, and Burkina Faso, respectively.

High Spatial Resolution Data by CNN with Accuracy Metric				
Country	Indicator	Percentage	Temporal	Spatio-Temporal
Rwanda	FCS	0.579	0.966	1.0
	HDDS	0.75	0.966	1.0
Tanzania	FCS	0.769	0.822	0.818
	HDDS	0.769	0.808	0.80
Burkina Faso	FCS	0.846	0.923	0.824
	HDDS	0.824	0.923	0.84

Table 4.5: Comparative Classification Results of CNN Model for High Spatial Resolution Data in Rwanda, Tanzania and Burkina Faso

Figures 4.6, 4.7 and 4.8 illustrate the spatial distribution comparison of actual versus predicted FCS classes(i.e Poor, Borderline, and Acceptance) using the CNN model with temporal split for Rwanda, Tanzania, and Burkina Faso, respectively. Each figure consists of three maps, the first map shows the predicted FCS classes, the second map displays the actual FCS classes, and the third map highlights the prediction errors of the CNN model across different regions. Across all three countries, the CNN model shows varying levels of accuracy in predicting FCS classes, with Rwanda displaying the closest alignment between predicted and actual values, followed by Burkina Faso, and then Tanzania. The geographic distribution of prediction errors indicates that while the model performs well overall in **Acceptance** and **Poor** classes, it faces challenges in accurately predicting **Borderline** regions.

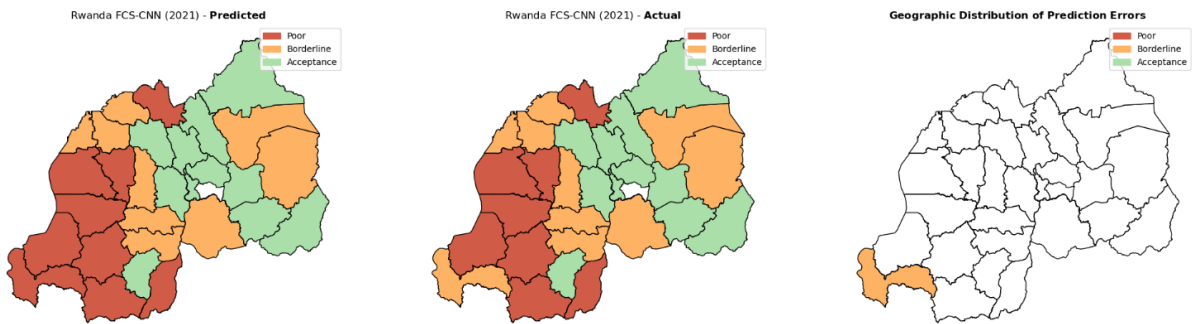


Figure 4.6: Spatial Distribution Comparison of **Actual vs. Predicted Classes** of FCS by Districts for Rwanda with High Spatial Data by CNN Model using Temporal Split

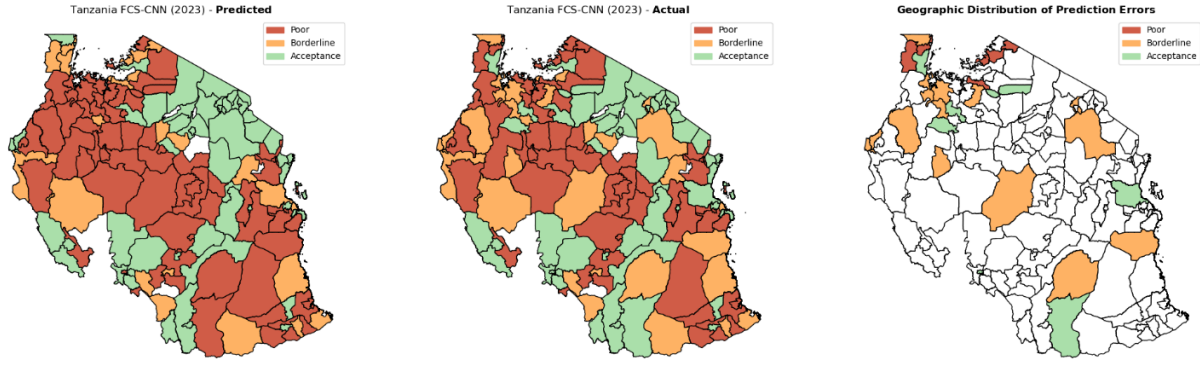


Figure 4.7: Spatial Distribution Comparison of **Actual vs. Predicted** Classes of FCS by Districts for Tanzania with High Spatial Data by CNN Model using Temporal Split

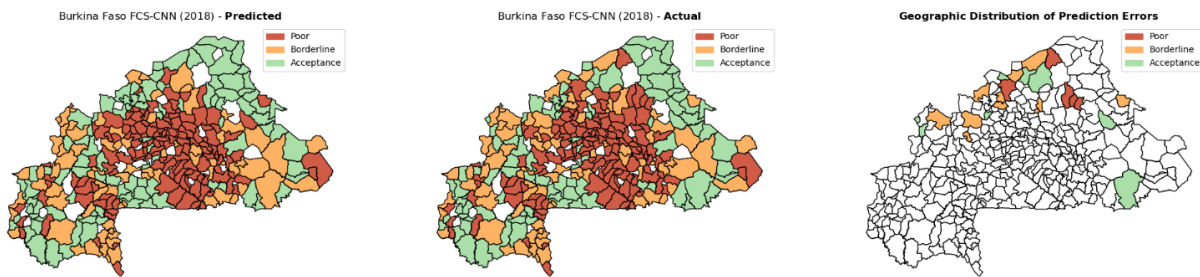


Figure 4.8: Spatial Distribution Comparison of **Actual vs. Predicted** Classes of FCS for Burkina Faso by Commune with High Spatial Data by CNN Model using Temporal Split

Conjunctural and Spatial Data

The Table 4.6 illustrate the performance of RF on classify the classes of FCS and HDDS for Conjunctural and Spatial data by using Robust Scaler across all the countries. The results demonstrate the variability of RF's performance based on data-splitting techniques, indicators, and countries. For instance, in Tanzania, the RF model reached an accuracy of **0.549** for FCS with the percentage split and **0.577** for HDDS with the spatio-temporal split. While in Burkina Faso, the highest accuracy was obtained using the temporal split for both FCS and HDDS, with values of **0.604** and **0.566**, respectively.

Conjunctural and Spatial Data by RF with Accuracy Metric				
Country	Indicator	Percentage	Temporal	Spatio-Temporal
Rwanda	FCS	0.739	0.724	0.87
	HDDS	0.739	0.586	0.72
Tanzania	FCS	0.549	0.462	0.49
	HDDS	0.511	0.405	0.577
Burkina Faso	FCS	0.533	0.604	0.494
	HDDS	0.493	0.566	0.444

Table 4.6: Comparative Classification Results of RF Model for Conjunctural and Spatial Data in Burkina Faso, Rwanda, and Tanzania

4.2 Discussion

This study aimed to evaluate the effectiveness and robustness of the FSPHD framework in assessing its generalizability for predicting FCS and HDDS using heterogeneous data sources from African countries, specifically Burkina Faso, Rwanda, and Tanzania. The framework incorporates various ML and DL models, including RF, LSTM, and CNN, which were tested on diverse data sources identified as key drivers of food security. Several experiments were conducted to gain insights into the performance of each model in predicting FCS and HDDS across different contexts. The results, as discussed in the previous section, demonstrate the significant variability in model performance based on data type, country, and data-splitting techniques, providing valuable insights into the suitability of each model for different scenarios.

Starting with the time series data evaluated by RF and LSTM models, the results highlight the overall performance of both models for this branch is not good, especially in Regression tasks. In the classification task, LSTM consistently outperformed RF, particularly when using temporal and spatio-temporal splits across all countries and indicators. However, regression performance varied significantly across different countries, indicators, and data-splitting techniques, making it difficult to identify a universally optimal split for this data category. Despite these challenges, LSTM remains the most suitable model for time-series data, particularly when temporal and spatio-temporal splits are applied, as these are crucial for capturing the inherent patterns and trends. However, it is important to note that even these splits did not always yield optimal results, likely due to the limited dataset size required to effectively train a DL model like LSTM or possibly because the features in this category are not sufficiently relevant. This suggests that further refinement is needed on the methodological side to improve model performance.

In the context of high spatial resolution data, the CNN model was applied, and the results demonstrate that its highest performance across all countries was achieved using temporal splits. In classification tasks, the model consistently outperformed regression tasks across all countries and indicators. Additionally, the analysis of conjunctural and spatial data using the RF model demonstrates that the model provides a promising results in classification tasks than in regression tasks across all countries. In these classification tasks, the model’s performance varied significantly depending on the data splits, indicators, and countries.

Overall, the framework shows strong performance in classification tasks across all the countries, particularly with the CNN model, which demonstrates the highest performance compared to other models. However, all models struggle to accurately predict the **borderline** class for both indicators. This is also observed in the study of [15] when their model struggled with predicting the ‘stressed’ a middle-ranked class using IPC phases classes. This problem could be attributed to the inherent complexities of calculating probabilities for middle-ranked classes. On the other hand, regression tasks that require the models to predict continuous outcomes yielded less favorable results due to the complexity and variability of the data, uneven distributions across countries, and the limitations of the data-splitting techniques. The lack of extensive, high-quality data further impacted the model’s ability to accurately capture trends and dependencies in regression tasks.

While the framework demonstrates strong performance on classification tasks especially by CNN mode, with other models providing promising results in their respective domains, there are still some operational limitations on the evaluation of the framework. Currently, the framework is configured to use the same data-splitting technique for both FCS and HDDS indicators within the same country, which may limit the evaluation of the framework’s performance. This limitation is evident not only in the time series data but also in the conjunctural and spatial

data, where in some cases, the highest performance for FCS is achieved with one split, while the best results for HDDS are obtained using a different split in the same country. For example, as shown in Table 4.1, in Rwanda the LSTM achieved R^2 of **0.409** with temporal splits but for HDDS it achieved the highest R^2 of **0.647** with percentage splits.

Furthermore, the current settings of the framework emphasizes selecting a generalized scaling method across all countries within each data category. However, the preliminary results indicate that this approach may not be ideal, as each country’s data has its unique distribution. The variation in data distribution particularly in time series data involving rainfall, temperature, and food prices data suggests that a one-size-fits-all scaling method is insufficient. For example, the distribution of data in Rwanda is unlikely to be the same as that of Tanzania, given the distinct social, economic, and environmental conditions in each country. As a result, applying a generalized scaling method across all countries can negatively impact model performance. This underscores the need for a more customized approach, where the choice of scaling method is adapted to the specific data distribution of each country, thereby optimizing model performance.

CHAPTER 5

Conclusion

5.1 Summary of Findings

The study successfully evaluated the generalizability and effectiveness of the FSPHD framework in predicting FCS and HDDS across Rwanda, Tanzania, and Burkina Faso using various ML and DL models. The CNN model demonstrated outstanding performance, particularly in classification tasks involving high spatial resolution data, with accuracies reaching approximately **97%**, **82%**, and **92%** for FCS, and **97%**, **81%**, and **92%** for HDDS in Rwanda, Tanzania, and Burkina Faso, respectively by using temporal split. Other models, such as LSTM for time-series data and RF for conjunctural and spatial data, also demonstrated promising results in classification tasks, with lower performance in regression tasks.

Overall, the framework proved somewhat generalized across the studied areas, though significant variability in model performance was observed depending on the data category, indicators, country, and data-splitting techniques. Importantly, the framework demonstrated stronger performance in classification tasks than regression tasks. This is particularly relevant in food security analysis since understanding whether a population’s food security status is poor, borderline, or acceptable is more important than knowing the exact numerical scores of FCS and HDDS. Thus, the superior classification performance underscores the framework’s ability to excel where it matters most by providing critical insights necessary for effective decision-making and intervention in food security contexts.

5.2 Limitations

Despite the promising results, the study faced several limitations. A key challenge was the quality and availability of household survey data used as ground truth. The dataset was insufficient for training DL models like LSTM, which typically require large volumes of data. Additionally, the data was not available at lower administrative levels, limiting the granularity of the analysis. For example, in Rwanda, the data was only available at the district level, covering 30 districts, rather than at the sector level, which would have provided finer granularity closer to household clusters. This lack of detailed data likely impacted the model’s performance, as more granular data could have led to better predictive accuracy. Another limitation was the use of FCS thresholds that were not fully aligned with WFP guidelines, as the classes were set using a percentile-based approach. This misalignment may have contributed to the observed difficulties in accurately predicting the **borderline** class, which was often misclassified as either **poor** or **acceptance** across all models and countries.

Moreover, the heterogeneity of the data in this study presented significant challenges. The diverse datasets, originating from various sources and consisting of different data types, required extensive preprocessing to ensure consistency and reliability. This process was time-consuming and resource-intensive, limiting our ability to use certain high-resolution data, such as NDVI for Tanzania, which required computational resources beyond our capacity, leading us to opt for lower-resolution alternatives.

In addition to these data-related challenges, several limitations were also observed within the framework itself. These include the use of a generalized scaling method for each data category across all studied countries, which likely impacted model performance, as each country has a unique data distribution. Additionally, applying the same data-splitting techniques for both FCS and HDDS indicators within the same country limited the framework’s ability to optimize model performance.

Moreover, the framework encountered significant limitations related to the data-splitting techniques used across different data categories. To achieve optimal performance, different data-splitting techniques were required for various models; for instance, temporal splits were most effective for the CNN model with high spatial data, while percentage splits worked best for the Random Forest model with conjunctural and spatial data. However, this inconsistency in data-splitting strategies posed a major challenge for implementing feature fusion, which relies on uniform data-splitting across all branches to ensure proper alignment of data shapes. Because the best features for each model required different data-splitting methods, achieving the necessary alignment for feature fusion was impossible. Therefore, we could not fully leverage the combined strengths of the models, potentially limiting the overall accuracy of predictions for FCS and HDDS by integrating data from different branches.

5.3 Future Work

This study has highlighted several areas for improvement within the FSPHD framework that could enhance its effectiveness and robustness across different contexts. A key area for future research is addressing the limitations of feature fusion. Specifically, there is a need for consistent data-splitting techniques that ensure strong model performance across different branches, thereby enabling the successful implementation of feature fusion.

Additionally, exploring country-specific scaling methods could address variability in data distributions, leading to more accurate predictions. Beyond these improvements, expanding the framework to incorporate diverse study areas across the globe, rather than focusing only on Africa, could further generalize its applicability. Since food insecurity is a global issue, applying the framework to different regions with varying socio-economic and environmental conditions would provide a broader understanding and enable more universally effective solutions.

Future research should also focus on strategies to increase the amount and quality of training data, which would help improve the performance of DL models, especially in regression tasks. Transfer learning could further strengthen model performance by applying knowledge from similar domains. Moreover, integrating additional data sources, such as those related to fisheries and livestock which is often overlooked in favor of agricultural data could provide a more comprehensive understanding of food security. Finally, adding real-time economic indicators and social media data could provide a better context and make the framework even better at predicting food security outcomes.

Bibliography

- [1] Hugo Deléglise et al. “Food security prediction from heterogeneous data combining machine and deep learning methods”. In: *Expert Systems with Applications* 190 (2022), p. 116189.
- [2] Food and Agriculture Organization of the United Nations (FAO). *Food Security*. Accessed: 14 March 2024. Rome, Italy, 2006. URL: http://fao.org/fileadmin/templates/faoitally/documents/pdf/pdf_Food_Security_Coecept_Note.pdf.
- [3] Food and Agriculture Organization. *World Food Summit: Rome Declaration on World Food Security*. 1996. URL: <http://www.fao.org/3/w3613e/w3613e00.htm>.
- [4] United Nations. *Transforming our world: the 2030 Agenda for Sustainable Development*. 2015. URL: <https://sustainabledevelopment.un.org/post2015/transformingourworld>.
- [5] United Nations. *Goal 2: Zero Hunger*. Accessed: 2024-03-28. URL: <https://www.un.org/sustainabledevelopment/hunger/>.
- [6] World Health Organization. *Malnutrition*. [Accessed: 10-April-2024]. 2021. URL: <https://www.who.int/news-room/fact-sheets/detail/malnutrition>.
- [7] UNICEF. *Levels and Trends in Child Malnutrition*. [Accessed: 10-April-2024]. 2023. URL: <https://data.unicef.org/wp-content/uploads/2023/05/JME-2023-Levels-and-trends-in-child-malnutrition.pdf>.
- [8] FSIN. *The Global Report on Food Crises 2023*. Accessed: 2024-03-25. 2023. URL: <https://www.fsinplatform.org/sites/default/files/resources/files/GRFC2023-compressed.pdf>.
- [9] FEWS NET. *About FEWS NET*. Accessed: 2024-05-29. 2021. URL: <https://fews.net>.
- [10] Global Information and Early Warning System (GIEWS). *Global Information and Early Warning System*. Accessed: 2024-05-29. 2021. URL: <https://www.fao.org/giews>.
- [11] International Institute for Sustainable Development. *World Population to Reach 9.9 Billion by 2050*. Accessed: 2024-03-25. 2021. URL: <https://sdg.iisd.org/news/world-population-to-reach-9-9-billion-by-2050/>.
- [12] Nouredine Jarray, Ali Ben Abbes, and Imed Riadh Farah. “Machine learning for food security: current status, challenges, and future perspectives”. In: *Artificial Intelligence Review* 56.6 (2023), pp. 3853–3876. DOI: 10.1007/s10462-023-10617-x. URL: <https://link.springer.com/article/10.1007/s10462-023-10617-x>.
- [13] Erin C. Lentz et al. “A data-driven approach improves food insecurity crisis prediction”. In: *World Development* 122 (2019), pp. 399–409. DOI: 10.1016/j.worlddev.2019.06.008.
- [14] Giulia Martini et al. “Machine learning can guide food security efforts when primary data are not available”. In: *Nature Food* 3 (2022), pp. 716–728.
- [15] Wesley van der Heijden et al. *Combining open data and machine learning to predict food security in Ethiopia*. Information Systems & Operations Management Research Group: Operations Research, Contribution to conference. 2018.
- [16] Food and Agriculture Organization of the United Nations. *Rwanda Country Profile*. Accessed: 2024-05-30. 2021. URL: <http://www.fao.org/countryprofiles/index/en/?iso3=RWA>.

- [17] The World Bank. *Rwanda Economic Update: Weathering the Storm*. Accessed: 2024-05-30. 2020. URL: <https://www.worldbank.org/en/country/rwanda/publication/rwanda-economic-update>.
- [18] M. E. Shongwe et al. “Projected Changes in Mean and Extreme Precipitation in Africa under Global Warming. Part II: East Africa”. In: *Journal of Climate* 24.14 (2011), pp. 3718–3733. DOI: 10.1175/2010JCLI2883.1.
- [19] Ahmed Gharib Khamis et al. “The influence of dietary diversity on the nutritional status of children between 6 and 23 months of age in Tanzania”. In: *BMC Pediatrics* 19.518 (2019). Open access, Published: 28 December 2019. DOI: 10.1186/s12887-019-1897-5. URL: <https://doi.org/10.1186/s12887-019-1897-5>.
- [20] Marco d’Errico, Donato Romano, and Rebecca Pietrelli. “Household resilience to food insecurity: evidence from Tanzania and Uganda”. In: *Food Security* 10 (2018). Published: 18 July 2018, pp. 1033–1054. DOI: 10.1007/s12571-018-0820-5. URL: <https://doi.org/10.1007/s12571-018-0820-5>.
- [21] World Bank. *Burkina Faso Overview: Development news, research, data*. Accessed: 2024-04-29. 2023. URL: <https://www.worldbank.org/en/country/burkinafaso/overview>.
- [22] World Food Programme. *WFP in Burkina Faso*. Accessed: 2024-04-29. 2024. URL: <https://www.wfp.org/countries/burkina-faso>.
- [23] World Food Programme. *Food Consumption Analysis*. Accessed: 2024-03-14. 2008. URL: <https://resources.vam.wfp.org/data-analysis/quantitative/food-security/food-consumption-score>.
- [24] National Institute of Statistics of Rwanda (NISR). *Rwanda - Comprehensive Food Security and Vulnerability Analysis (CFSVA)*. Accessed: 2024-03-16. 2006. URL: <https://microdata.statistics.gov.rw/index.php/catalog/26>.
- [25] National Institute of Statistics of Rwanda (NISR). *RWANDA - Comprehensive Food Security and Vulnerability Analysis and Nutrition Survey 2012*. Accessed: 2024-03-16. 2012. URL: <https://microdata.statistics.gov.rw/index.php/catalog/69>.
- [26] National Institute of Statistics of Rwanda (NISR). *Rwanda - Comprehensive Food Security and Vulnerability Analysis (CFSVA)*. Accessed: 2024-03-16. 2015. URL: <https://microdata.statistics.gov.rw/index.php/catalog/70>.
- [27] National Institute of Statistics of Rwanda (NISR). *Rwanda - Comprehensive Food Security and Vulnerability Analysis 2018*. Accessed: 2024-03-16. 2018. URL: <https://microdata.statistics.gov.rw/index.php/catalog/91>.
- [28] National Institute of Statistics of Rwanda (NISR). *Rwanda - Comprehensive Food Security and Vulnerability Analysis 2021*. Accessed: 2024-03-16. 2021. URL: <https://microdata.statistics.gov.rw/index.php/catalog/106>.
- [29] National Bureau of Statistics. *National Panel Survey 2010-2011, Wave 2*. Accessed: 2024-03-17. 2011. DOI: 10.48529/jm20-c742. URL: <https://doi.org/10.48529/jm20-c742>.
- [30] National Bureau of Statistics. *National Panel Survey 2012-2013, Wave 3*. Accessed: 2024-03-17. 2013. DOI: 10.48529/7vqv-5f71. URL: <https://doi.org/10.48529/7vqv-5f71>.
- [31] National Bureau of Statistics. *National Panel Survey 2014-2015, Wave 4*. Accessed: 2024-03-17. 2015. DOI: 10.48529/y3qj-d018. URL: <https://doi.org/10.48529/y3qj-d018>.
- [32] National Bureau of Statistics. *National Panel Survey 2019-2020 - Extended Panel with Sex Disaggregated Data*. Accessed: 2024-03-17. 2020. DOI: 10.48529/0y7d-1v78. URL: <https://doi.org/10.48529/0y7d-1v78>.

- [33] National Bureau of Statistics. *National Panel Survey 2020-21, Wave 5*. Accessed: 2024-03-17. 2022. URL: <https://microdata.worldbank.org/index.php/catalog/5639/get-microdata>.
- [34] National Bureau of Statistics. *High Frequency Welfare Monitoring Phone Survey 2021-2024*. Accessed: 2024-03-17. 2024. URL: <https://microdata.worldbank.org/index.php/catalog/4542>.
- [35] National Centers for Environmental Information (NCEI). *Climate Data Records: Geostationary IR Channel Brightness Temperature*. Accessed: 2024-05-10. 2024. URL: <https://www.ncei.noaa.gov/products/climate-data-records/geostationary-IR-channel-brightness-temperature>.
- [36] Copernicus Climate Data Store. *ERA5 Dataset: Reanalysis Single Levels*. Accessed: 2024-04-25. 2024. URL: <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=overview>.
- [37] Copernicus Climate Data Store. *API Documentation*. Accessed: 2024-04-25. 2024. URL: <https://cds.climate.copernicus.eu/api-how-to>.
- [38] World Food Programme (WFP). *WFP Food Prices for United Republic of Tanzania*. Accessed: 2024-04-18. 2024. URL: <https://data.humdata.org/dataset/wfp-food-prices-for-united-republic-of-tanzania>.
- [39] World Food Programme (WFP). *WFP Food Prices for Rwanda*. Accessed: 2024-04-18. 2024. URL: <https://data.humdata.org/dataset/wfp-food-prices-for-rwanda>.
- [40] GHSL - Global Human Settlement Layer. *Population Data*. <https://human-settlement.emergency.copernicus.eu/download.php?ds=pop>. Accessed: 2024-06-17. 2024. URL: <https://human-settlement.emergency.copernicus.eu/download.php?ds=pop>.
- [41] World Bank. *World Bank GDP Growth Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG>.
- [42] World Bank. *World Bank GDP per Capita Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG>.
- [43] World Bank. *World Bank Trade Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/NE.TRD.GNFS.ZS>.
- [44] World Bank. *World Bank National Expenditure Growth Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/NE.DAB.TOTL.ZS>.
- [45] World Bank. *World Bank Household Final Consumption Expenditure Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/NE.CON.PRVT.ZS>.
- [46] World Bank. *World Bank Inflation Consumer Prices Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG>.
- [47] World Bank. *World Bank Population Growth Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/SP.POP.GROW>.
- [48] World Bank. *World Bank Unemployment Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/SL.UEM.TOTL.ZS>.
- [49] World Bank. *World Bank FDI Outflows Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/BM.KLT.DINV.WD.GD.ZS>.
- [50] World Bank. *World Bank FDI Inflows Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/BX.KLT.DINV.WD.GD.ZS>.

- [51] World Bank. *World Bank Merchandise Trade Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/TG.VAL.TOTL.GD.ZS>.
- [52] World Bank. *World Bank Military Expenditure Data for Tanzania and Rwanda*. Accessed: 2024-04-30. 2024. URL: <https://data.worldbank.org/indicator/MS.MIL.XPND.GD.ZS>.
- [53] K. Didan. *MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061*. Accessed 2024-05-03. 2021. URL: <https://doi.org/10.5067/MODIS/MOD13Q1.061>.
- [54] K. Didan. *MODIS/Terra Vegetation Indices 16-Day L3 Global 500m SIN Grid V061*. <https://doi.org/10.5067/MODIS/MOD13A1.061>. Accessed 2024-05-15. 2021.
- [55] AppEEARS Team. *Application for Extracting and Exploring Analysis Ready Samples (AppEEARS)*. Version 3.53. Accessed May 3, 2024. 2024. URL: <https://appeears.earthdatacloud.nasa.gov>.
- [56] OpenStreetMap contributors. *OpenStreetMap*. Accessed: 2024-05-24. 2024. URL: <https://www.openstreetmap.org>.
- [57] ACLED. *Armed Conflict Location & Event Data Project (ACLED) Data*. Accessed: 2024-05-04. 2024. URL: <https://acleddata.com/>.
- [58] FAO and IIASA. *Harmonized World Soil Database (version 2.0)*. Accessed: 2024-05-30. 2024. URL: <https://www.fao.org/family-farming/detail/en/c/1635950/>.
- [59] NASA JPL. *NASA Shuttle Radar Topography Mission Global 3 arc second NetCDF*. Accessed: 2024-05-16. 2013. URL: https://doi.org/10.5067/MEaSURES/SRTM/SRTMGL3_NC.003.
- [60] Esri. *Sentinel-2 Land Cover Explorer*. Accessed: 2024-05-20. 2022. URL: <https://livingatlas.arcgis.com/landcoverexplorer/#mapCenter=-3.28600%2C31.34000%2C3&mode=step&timeExtent=2017%2C2021&year=2022>.
- [61] Sepp Hochreiter and Jürgen Schmidhuber. “Long Short-Term Memory”. In: *Neural Computation* 9.8 (1997), pp. 1735–1780. DOI: 10.1162/neco.1997.9.8.1735. URL: <https://doi.org/10.1162/neco.1997.9.8.1735>.
- [62] K. Bhosle and V. Musande. “Evaluation of Deep Learning CNN Model for Land Use Land Cover Classification and Crop Identification Using Hyperspectral Remote Sensing Images”. In: *Journal of the Indian Society of Remote Sensing* 47 (2019), pp. 1949–1958. DOI: 10.1007/s12524-019-01041-2. URL: <https://doi.org/10.1007/s12524-019-01041-2>.

APPENDIX A

Internship Progress

The internship started on **11th March 2024** and ended on **9th September 2024**. It was conducted on-site at Maison de la Télédétection (MTD) in Montpellier, France, under Centre de coopération Internationale en Recherche Agronomique pour le Développement (CIRAD) within the Department of Environments and Societies (ES). I worked in the research unit of the Land, Environment, Remote Sensing, and Spatial Information (UMR -TETIS).

During the internship, regular meetings were held with the supervisors. These meetings occurred every week and sometimes after two weeks. The purpose of these meetings was to discuss the progress of the project, address any issues, and set objectives for the project's next phases. Below is a table summarizing the key meetings and presentations that took place during the internship. This includes the date of each meeting, the attendees, and the objectives or topics discussed.

Date	Attendees	Type of Meeting	Main Objectives/ Discussion
11/03/2024	Asha Mehtab Simon	Initial Project Meeting	- Overview of the research project on predicting food security indicators (FCS and HDDS) using heterogeneous data sources for Tanzania and Rwanda. - Getting Familiar with the previous work that develops FSPHD Next Steps: - Doing a Literature Review on the paper that develops FSPHD - Begin collecting survey data for FCS and HDDS calculations for Rwanda and Tanzania from LSMS and Rhomiz
20/03/2024	Asha Roberto Simon	Progress Meeting	- Provided an update on data collection for Tanzania and Rwanda. - Discuss some challenges faced (such as the unavailability of data for computing those scores) in the dataset and get some suggestions on how to deal with those challenges from my supervisors Next Steps: - Start Computing the FCS and HDDS for those available datasets. - Looking for more household survey dataset - Provide Spatial Distribution of the Data on the Map
28/03/2024	Asha Mehtab Roberto Simon	Progress Meeting	- Presenting the additional dataset obtained and how the FCS and HDDS will be calculated. - Discussing how to deal with the dataset that misses some information for calculating HDDS Next Steps: - Use FCS data to compute HDDS - Reconcile the administrative name such as province, district, and others to match with those available for the existing shapefile of administrative boundaries of each country - Show the Spatial distribution of the score on the Map based on the WFP thresholds to set for FCG such as Poor, Borderline and Acceptance
17/04/2024	Asha Mehtab Roberto Simon	Progress Meeting	- Presenting the Map that shows Spatial Distribution and each number of samples per district. - Discussing the issues of class imbalance shown on the map that could affect our predictions Next Steps: - Explore the guideline to create other FCG thresholds provided by WFP to see if we can get a balance of classes - Start collecting feature data such as rainfall, temperature, NDVI and others.
22/05/2024	Asha Jérémy Mehtab Roberto Simon	Progress Meeting	- Discussing the new method used to create the FCG thresholds (using a Percentile-based Approach) to get a more balanced class - Presenting some of the Features data collected, the sources, resolution, and preprocessing done - Discussing the structure of Manuscript Writing Next Steps - Continue with the remaining feature data collection and preprocessing and start writing the manuscript

03/06/2024	Asha Nacéra Mehtab Roberto Simon	Mid- Presentation of my internship	- This was the presentation to give a general idea about my project such as the context, main objectives, what I have done so far, and what I am expecting. - Discussing with Nacéra about the structure of my manuscripts and get some suggestions from her
06/06/2024	Asha Mehtab Roberto Simon	Progress Meeting	- Discussing about the finalization of all feature data collection and preprocessing Next Steps: - Working together with Mehtab for model development (model training and testing)
13/06/2024	All Interns (TETIS) Some Staff (TETIS)	Stagiairothon	This is the one-day presentation dedicated to all interns at TETIS. In this presentation, every intern is given 5min to present his/her project and 5min for questions and answers.
24/06/2024	Asha Mehtab Roberto Simon	Progress Meeting	- Presenting the results of Model Training and Evaluation on the Time-series Data using LSTM and RF - Discussing some challenges faced in dealing with raster data for model training and getting some suggestions from supervisors on how we can deal with that Next Steps: - Incorporating the classification tasks for the model training and visualizing the results on a map - Working with model training and evaluation on Raster Data using CNN - Improving the model of LSTM to get better results
17/07/2024	Asha Jérémy Mehtab Roberto	Progress Meeting	- Presenting the results of LSTM and CNN for both regression and classification tasks and displaying the results on a map and some graphs of metrics - Discussing about data splitting technique used (percentage and temporary splits) Next Steps: - Try different strategies for data splitting for training and testing - Performing feature fusion of Data from LSTM, CNN and data from conjunctural and spatial data to get the final predictions using RF
29/07/2024	Asha Simon	Progress Meeting	- Discussing the results of the LSTM and RF on time-series data and CNN based on different data-splitting techniques Next Steps: - Using RF for conjunctural and spatial data and evaluating them with different scaling methods and techniques - Implement the feature fusion to integrate all four data categories and make predictions by using RF
19/08/2024	Asha Simon	Progress Meeting	Discussion about the experiment results of all the models with their corresponding data categories and the main findings of the whole project.

APPENDIX B

Illustration of Dietary Diversity Pattern

To address the class imbalance problem in Rwanda and Tanzania. We use the guidelines of the WPF to find the proper thresholds for our study areas. This involves checking the dietary diversity pattern of each country and the consumption of sugar and oil as explained in this guideline [23]. It was suggested that if the area under consideration shows widespread of consumption of sugar and oil, which nearly all people in that area consume sugar and oil daily (in 7 days), it would impact the FCS score and therefore the new thresholds should be $[0 - 28]$, $[28.5 - 42]$, and $[\geq 42]$

The graphs illustrate the dietary patterns for both Rwanda B.1(a)B.1(b) and Tanzania B.2(a)B.2(b) for the survey data. However, the dietary pattern shows that there are no significant issues related to sugar and oil consumption and also the pattern aligns well with the default thresholds of $[0 - 21]$, $[21.5 - 35]$, and $[\geq 35]$ representing poor, borderline, and acceptance respectively. This suggests that the default thresholds are appropriate for assessing dietary quality in our study area. Since this thresholds give a class imbalance the only solution was to use the percentile-based approach.

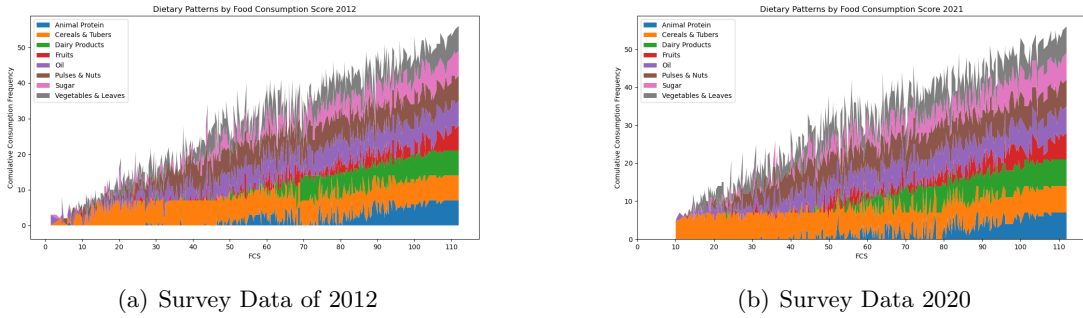


Figure B.1: Dietary Diversity Pattern of Rwanda

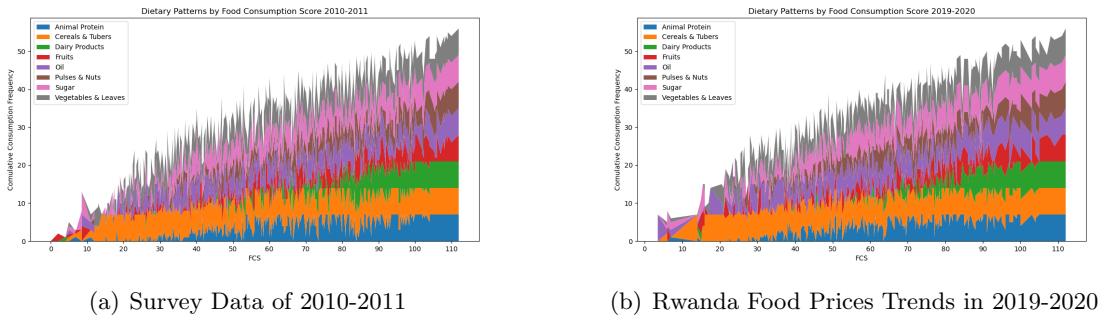


Figure B.2: Dietary Diversity Pattern of Tanzania

APPENDIX C

Illustration of Time series Data Distribution

The category of Time series data involves data with different magnitudes. For instance, food price values and rainfall readings represent different kinds of data and may exhibit different distributions as illustrated in Figure C.1 where rainfall data shows a highly skewed distribution, maize data price shows a moderately skewed distribution, and smooth brightness temperature displays a roughly normal distribution. Therefore using these datasets without appropriate normalization could lead to suboptimal prediction results. Therefore we applied all three scaling methods techniques; the standard scaler, min-max scaler, and robust to evaluate the best among them.

To use these techniques we analyze the distribution of each attribute in this category and use different scaling methods based on its data distribution. For example Figure C.2 of the box plot shows the distribution of maize prices across different months and chooses an appropriate scaling method. In that case, Standard Scaler was applied since the data shows the normal distribution, and Figure C.3 shows the distribution of the data after scaling.

This approach was applied to all the data in this category of time series. However, it did not improve predictive performance. Then we decided to conduct several experiments that will be explained in next appendix D to find a more generalized scaling method for all the time series data regardless of the distribution that will show better performance for both regression and classification across all the countries.

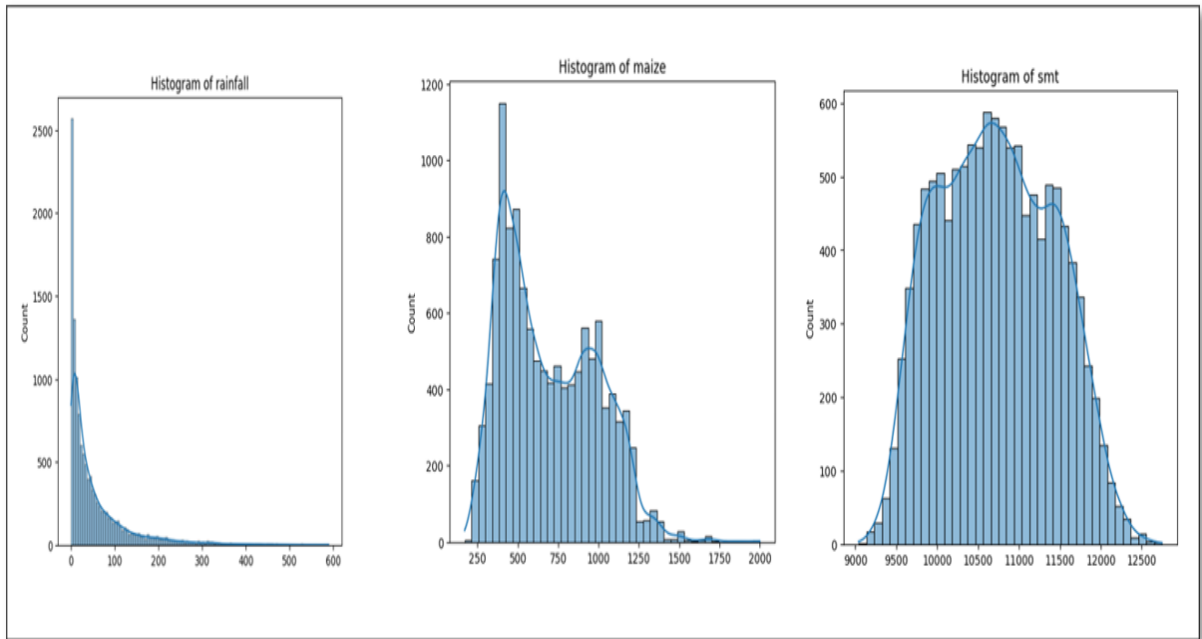


Figure C.1: Some of Time Series Data Distribution For Tanzania

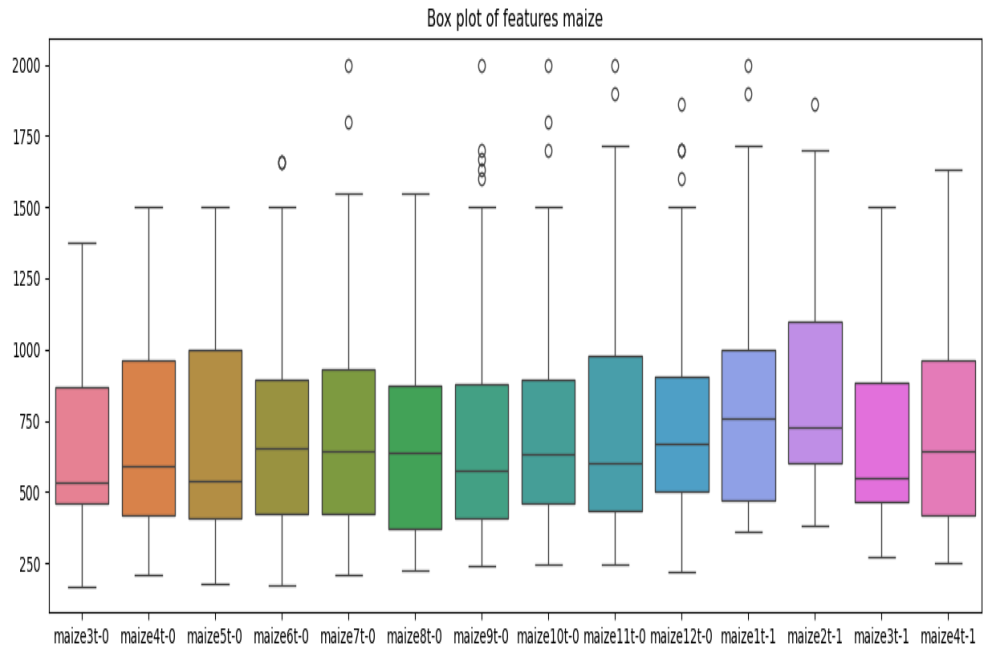


Figure C.2: Maize Data Distribution For Each Month

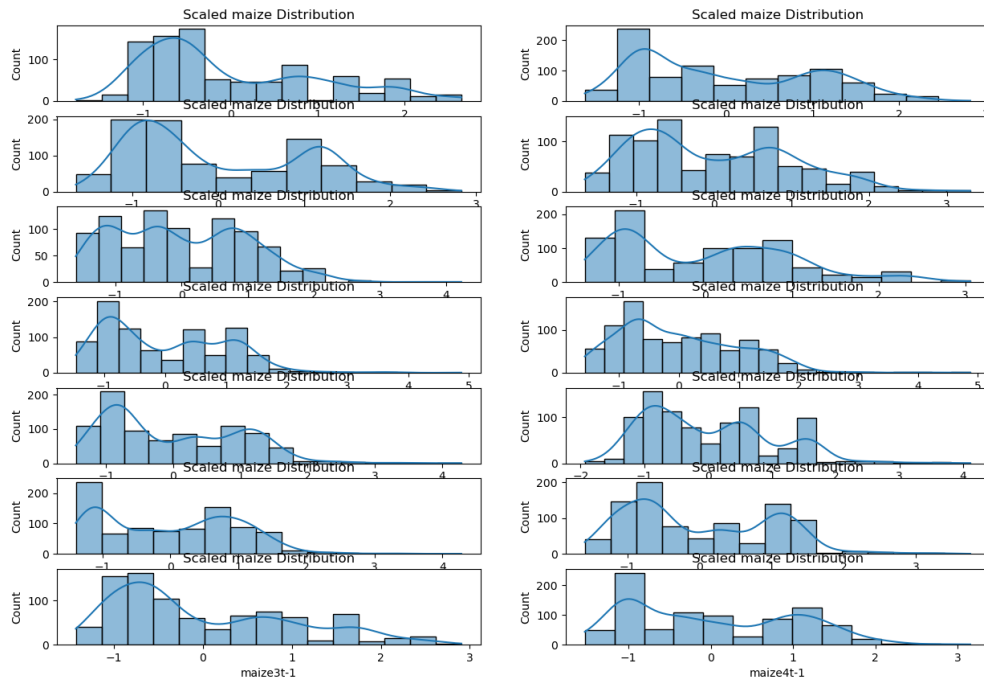


Figure C.3: Scaled Data of Maize

APPENDIX D

Preliminary Results for Time series Data

This appendix presents the preliminary results from the time series data experiments using both the Random Forest (RF) and Long Short-Term Memory (LSTM) models. The objective of these experiments was to determine a generalized scaling method that could effectively normalize diverse types of time series data across different countries, enhancing the predictive performance of the models for both regression and classification tasks.

Three scaling methods were tested including Standard, Min-Max, and Robust scaler across all the data splitting techniques and countries. Among the different techniques tested, the Min-Max Scaler seems to be the most effective, particularly when configured to scale features to a range of $(-1, 1)$. The Min-Max Scaler showed superior performance in enhancing the accuracy and consistency of both regression and classification tasks by using LSTM. However, Robust Scaler seems to perform better in RF models across various tasks and countries.

Despite the Robust Scaler's ability to perform better in the RF model, the Min-Max Scaler was chosen as a generalized scaler method across both models and all the countries. This is because of its ability to maintain the relative relationships between data points, which is crucial for time series data. The Min-Max Scaler preserves the original data shape while adjusting the range. This preservation of data integrity is important in time series data, where the pattern and seasonality are critical for accurate predictions.

The following tables display the performance of each model across all scalers, countries, indicators, and data splits, which collectively support the decision to choose the Min-Max Scaler for time series data.

RANDOM FOREST - PERCENTAGE SPLIT					
Country	Task	Indicator	Standard	Min-max	Robust
Rwanda	Regression	FCS	0.59	0.346	0.654
		HDDS	0.637	0.257	0.675
	Classification	FCS	0.696	0.609	0.552
		HDDS	0.652	0.609	0.783
Tanzania	Regression	FCS	0.183	0.234	0.308
		HDDS	0.268	0.121	0.277
	Classification	FCS	0.541	0.511	0.586
		HDDS	0.526	0.481	0.549
Burkina Faso	Regression	FCS	0.284	0.233	0.199
		HDDS	0.263	0.177	0.242
	Classification	FCS	0.496	0.452	0.465
		HDDS	0.446	0.459	0.439

RANDOM FOREST - TEMPORAL SPLIT					
Country	Task	Indicator	Standard	Min-max	Robust
Rwanda	Regression	FCS	0.393	0.351	0.259
		HDDS	-0.195	-0.17	-0.212
	Classification	FCS	0.517	0.517	0.586
		HDDS	0.586	0.586	0.552
Tanzania	Regression	FCS	0.048	0.038	0.061
		HDDS	-0.108	-0.122	-0.122
	Classification	FCS	0.424	0.468	0.468
		HDDS	0.449	0.411	0.386
Burkina Faso	Regression	FCS	0.001	-0.023	0.009
		HDDS	0.094	0.077	0.046
	Classification	FCS	0.412	0.418	0.431
		HDDS	0.494	0.491	0.487

RANDOM FOREST - SPATIO-TEMPORAL SPLIT					
COUNTRY	Task	Indicator	Standard	Min-Max	Robust
Rwanda	Regression	FCS	0.047	0.286	0.385
		HDDS	0.552	0.602	0.699
	Classification	FCS	0.6	0.6	0.565
		HDDS	0.56	0.56	0.64
Tanzania	Regression	FCS	0.092	0.264	0.32
		HDDS	0.173	0.225	0.169
	Classification	FCS	0.508	0.444	0.542
		HDDS	0.5	0.582	0.486
Burkina Faso	Regression	FCS	0.129	0.101	0.278
		HDDS	0.097	0.184	0.147
	Classification	FCS	0.42	0.43	0.503
		HDDS	0.459	0.463	0.464

LSTM PERCENTAGE SPLIT					
Country	Task	Indicator	Standard	Min-max	Robust
Rwanda	Regression	FCS	-0.906	0.282	-1.291
		HDDS	0.62	0.647	0.807
	Classification	FCS	0.609	0.552	0.652
		HDDS	0.739	0.609	0.565
Tanzania	Regression	FCS	0.30	0.25	0.253
		HDDS	0.08	0.217	0.184
	Classification	FCS	0.75	0.541	0.504
		HDDS	0.496	0.541	0.617
Burkina Faso	Regression	FCS	0.079	0.151	0.154
		HDDS	0.189	0.225	0.145
	Classification	FCS	0.476	0.498	0.498
		HDDS	0.50	0.493	0.504

LSTM TEMPORAL SPLIT					
Country	Task	Indicator	Standard	Min-max	Robust
Rwanda	Regression	FCS	-0.9243	0.409	-1.129
		HDDS	0.59	0.321	-0.247
	Classification	FCS	0.724	0.69	0.655
		HDDS	0.655	0.586	0.586
Tanzania	Regression	FCS	0.06	0.043	0.086
		HDDS	-0.126	0.014	-0.085
	Classification	FCS	0.5	0.513	0.506
		HDDS	0.449	0.437	0.449
Burkina Faso	Regression	FCS	0.041	0.018	0.074
		HDDS	0.187	0.194	0.221
	Classification	FCS	0.421	0.516	0.478
		HDDS	0.509	0.519	0.509

LSTM SPATIO-TEMPORAL SPLIT					
COUNTRY	Task	Indicator	Standard	Min-Max	Robust
Rwanda	Regression	FCS	-4.243	0.036	-0.724
		HDDS	0.428	0.518	0.698
	Classification	FCS	0.58	0.84	0.6
		HDDS	0.72	0.76	0.48
Tanzania	Regression	FCS	0.073	0.021	0.278
		HDDS	0.126	0.088	0.136
	Classification	FCS	0.515	0.559	0.59
		HDDS	0.588	0.548	0.486
Burkina Faso	Regression	FCS	0.018	0.158	-0.031
		HDDS	0.148	0.231	0.191
	Classification	FCS	0.448	0.515	0.474
		HDDS	0.509	0.523	0.47

APPENDIX E

Preliminary Results for Conjunctural and Spatial Data

This appendix presents the results of preliminary experiments conducted on Conjunctural and Spatial Data using Random Forest (RF) with three different data splitting techniques. The objective of these experiments was to identify the best scaling method for normalizing the data across all study areas and to find a method that generalizes well across diverse regions.

The results indicate that while certain scaling methods performed well for specific countries, they did not consistently generalize across all study areas. For example, the Standard Scaler method showed strong performance in Burkina Faso but did not perform as well in the other regions. Conversely, the Robust Scaler demonstrated better generalization across all three study areas, performing well in most cases for both regression and classification tasks. As a result, the Robust Scaler was selected as the preferred method for normalizing the data in this category. The evaluation metric were R^2 for Regression and Accuracy for Classification.

RF PERCENTAGE SPLIT					
Country	Task	Indicator	Standard	Min-max	Robust
Rwanda	Regression	FCS	0.534	0.712	0.841
		HDDS	0.745	0.694	0.807
	Classification	FCS	0.609	0.478	0.696
		HDDS	0.696	0.565	0.87
Tanzania	Regression	FCS	0.201	0.179	0.263
		HDDS	0.293	0.232	0.177
	Classification	FCS	0.534	0.579	0.489
		HDDS	0.564	0.534	0.571
Burkina Faso	Regression	FCS	0.347	0.314	0.325
		HDDS	0.295	0.282	0.256
	Classification	FCS	0.513	0.493	0.522
		HDDS	0.478	0.55	0.507
RF TEMPORAL SPLIT					
Country	Task	Indicator	Standard	Min-max	Robust
Rwanda	Regression	FCS	0.618	0.597	0.574
		HDDS	0.558	0.535	0.455
	Classification	FCS	0.724	0.724	0.724
		HDDS	0.552	0.586	0.586
Tanzania	Regression	FCS	0.002	-0.048	0.003
		HDDS	-0.118	-0.096	-0.094
	Classification	FCS	0.462	0.456	0.43
		HDDS	0.405	0.418	0.43
Burkina Faso	Regression	FCS	0.222	0.248	0.271
		HDDS	0.293	0.301	0.288
	Classification	FCS	0.597	0.588	0.579
		HDDS	0.569	0.579	0.544
RF SPATIO-TEMPORAL SPLIT					
COUNTRY	Task	indicator	Standard	Min-Max	Robust
Rwanda	Regression	FCS	0.392	0.423	0.548
		HDDS	0.525	0.68	0.768
	Classification	FCS	0.72	0.792	0.72
		HDDS	0.6	0.44	0.652
Tanzania	Regression	FCS	0.029	0.138	0.146
		HDDS	0.327	0.274	0.2
	Classification	FCS	0.549	0.489	0.557
		HDDS	0.409	0.511	0.379
Burkina Faso	Regression	FCS	0.218	0.075	0.138
		HDDS	0.146	0.122	0.117
	Classification	FCS	0.456	0.453	0.404
		HDDS	0.468	0.456	0.45